

Homeostatic Synaptic Plasticity of Miniature Excitatory Postsynaptic Currents in Mouse Cortical Cultures Requires Neuronal Rab3A

Reviewed Preprint

v2 • October 9, 2024

Revised by authors

Reviewed Preprint

v1 • September 14, 2023

Andrew G Koesters , Mark M Rich, Kathrin L Engisch

Department of Pharmacology and Systems Physiology, University of Cincinnati College of Medicine, Cincinnati, USA • Department of Neuroscience, Cell Biology and Physiology, Boonshoft School of Medicine, Wright State University, Dayton, USA • Department of Neuroscience, Cell Biology and Physiology, Boonshoft School of Medicine and the College of Science and Mathematics, Wright State University, Dayton, USA

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Following prolonged activity blockade, amplitudes of miniature excitatory postsynaptic currents (mEPSCs) increase, a form of plasticity termed “homeostatic synaptic plasticity.” We previously showed that a presynaptic protein, the small GTPase Rab3A, is required for full expression of the increase in miniature endplate current amplitudes following prolonged blockade of action potential activity at the mouse neuromuscular junction in vivo (Wang et al., 2011), but it is unknown whether this form of Rab3A-dependent homeostatic plasticity shares any characteristics with central synapses. We show here that homeostatic synaptic plasticity of mEPSCs is impaired in mouse cortical neuron cultures prepared from Rab3A^{-/-} and mutant mice expressing a single point mutation of Rab3A, Rab3A Earlybird mice. To determine if Rab3A is involved in the well-established homeostatic increase in postsynaptic AMPA-type receptors (AMPA receptors), we performed a series of experiments in which electrophysiological recordings of mEPSCs and confocal imaging of synaptic AMPAR immunofluorescence were assessed within the same cultures. We found that Rab3A was required for the increase in synaptic AMPARs following prolonged activity blockade, but the increase in mEPSC amplitudes was not always accompanied by an increase in postsynaptic AMPAR levels, suggesting other factors may contribute. Finally, we demonstrate that Rab3A is acting in neurons because only selective loss of Rab3A in neurons, not glia, disrupted the homeostatic increase in mEPSC amplitudes. This is the first demonstration that neuronal Rab3A is required for homeostatic synaptic plasticity and that it does so partially through regulation of the surface expression of AMPA receptors.

eLife Assessment

This study presents **useful** findings on the role of the small GTPase Rab3A in homeostatic synaptic plasticity following activity suppression. While the study demonstrates that Rab3A is required for homeostatic scaling, the evidence supporting the model put forward by the authors is **incomplete**. The work will be of interest to the field of synaptic transmission and synaptic plasticity.

<https://doi.org/10.7554/eLife.90261.2.sa4>

Introduction

One of the most studied phenomena triggered by prolonged activity blockade is the increase in amplitudes of miniature excitatory postsynaptic currents (mEPSCs) in neurons. First demonstrated in cultures of dissociated cortical neurons (Turrigiano et al., 1998 [↗](#)) and spinal cord neurons (O'Brien et al., 1998 [↗](#)), the compensatory response to prolonged loss of activity has been dubbed “homeostatic synaptic plasticity” (Turrigiano and Nelson, 2004 [↗](#); Pozo and Goda, 2010 [↗](#)). These first two studies provided evidence that glutamate receptors were also increased after inactivity, via an increased response to exogenously applied glutamate (Turrigiano et al., 1998 [↗](#)) and increased immunofluorescent labeling of GluA1 receptors at synaptic sites (O'Brien et al., 1998 [↗](#)). The accompanying increase in synaptic AMPA receptors has been confirmed with immunofluorescence in multiple studies of hippocampal and cortical cultures treated with TTX (Wierenga et al., 2005 [↗](#); Hou et al., 2008 [↗](#); Ibata et al., 2008 [↗](#); Jakawich et al., 2010a [↗](#); Tatavarty et al., 2013 [↗](#); Xu and Pozzo-Miller, 2017 [↗](#); Dubes et al., 2022 [↗](#)). It is now well accepted that the homeostatic increase in mEPSC amplitudes in neurons is due to an increase in postsynaptic AMPA receptors.

In our previous work, we found that a TTX cuff applied for 48 hr around the sciatic nerve in mice led to an increase in the amplitude of the miniature endplate currents (mEPCs) recorded in tibialis anterior muscles. Surprisingly, and unlike central synapses, we could find no evidence that the acetylcholine receptors levels at the neuromuscular junction (NMJ) were increased in the TTX-cuff treated mice (Wang et al., 2005 [↗](#)). This result led us to search for presynaptic molecules that might homeostatically regulate the presynaptic quantum. In previous studies in chromaffin cells, we identified the small GTPase Rab3A, a presynaptic vesicle protein, as a regulator of synaptic vesicle fusion pore opening (Wang et al., 2008 [↗](#)), so we examined whether deletion of Rab3A (Rab3A^{-/-}) might prevent homeostatic upregulation of mEPC amplitude at the NMJ. The results were clear: the homeostatic increase in mEPC amplitude induced by the TTX cuff was strongly reduced in the Rab3A^{-/-} mouse, and was completely abolished in the Earlybird mutant (Rab3A^{Ebd/Ebd}), which has a single point mutation in Rab3A that causes a shift towards early awakening that is more dramatic than that of the Rab3A deletion mouse (Kapfhammer et al., 2002 [↗](#); Wang et al., 2011 [↗](#)). These results led us to conclude that the homeostatic increase in mEPC amplitude at the NMJ is a presynaptic phenomenon.

It has remained somewhat of a mystery what the role of Rab3A is in synaptic transmission. The Rab3A^{-/-} mouse has minimal phenotypic abnormalities, with evoked synaptic transmission and mEPSCs essentially normal in hippocampal slices (Geppert et al., 1994 [↗](#)). At the Rab3A^{-/-} NMJ, reductions in evoked transmission were detected, but only under conditions of reduced extracellular calcium (Coleman et al., 2007 [↗](#)). There are some modest changes in short term plasticity during repetitive stimulus trains, with increased depression observed in response to moderate frequencies in hippocampal slices (Geppert et al., 1994 [↗](#)) and at the mammalian NMJ

(Coleman and Bykhovskaia, 2009 [↗](#)), but increased facilitation in response to high frequency stimulation at the mammalian NMJ (Coleman and Bykhovskaia, 2010 [↗](#)). The most dramatic effect of loss of Rab3A is the abolishment of a presynaptic form of long-term potentiation (LTP) at the mossy fiber-CA3 synapse that is induced by 25 Hz stimulation in the presence of NMDA blockers (Weisskopf et al., 1994 [↗](#); Castillo et al., 1997 [↗](#)). Our result that prolonged inactivity-induced plasticity of mEPSC amplitude at the mammalian NMJ is disrupted with loss of function of Rab3A further cements a role for Rab3A in activity-dependent plasticity of synaptic strength. However, a crucial question that remains is whether the TTX-cuff induced effect on the amplitude of the spontaneous synaptic current at the NMJ shares underlying mechanisms with central synapses. We set out to answer this question for the more well-studied phenomenon of homeostatic plasticity of mEPSCs in dissociated mouse cortical neurons prepared from wild type, Rab3A^{-/-} mice and Rab3A^{Ebd/Ebd} mice treated with TTX for 48 hr. We show here that homeostatic synaptic plasticity of mEPSC amplitude was strongly reduced in cultures from Rab3A^{-/-} mice, and completely abolished in cultures from Rab3A^{Ebd/Ebd} mice, supporting a shared mechanism with the neuromuscular junction. Unexpectedly, the modest increase in GluA2 immunofluorescence we observed in wild type cultures was not present in Rab3A^{-/-} cultures. Loss of Rab3A in neurons, but not in glia, disrupted the increase in mEPSC amplitudes after TTX treatment. These results support the idea that the increase in mEPSC amplitude is due to more postsynaptic receptors, and that neuronal Rab3A is necessary for this regulation. However, we found that GluA2 receptors and mEPSC amplitudes did not always change in parallel. This mismatch suggests there may be contributors other than postsynaptic receptor numbers to the homeostatic plasticity of mEPSC amplitude.

Results

We previously reported that mixed cultures of cortical neurons and glia prepared from postnatal day 0-2 mouse pups responded to a block of action potential-mediated activity by a 48 hr TTX treatment with an increase in mEPSC amplitude (Hanes et al., 2020 [↗](#)). Here, we asked the question whether cortical cultures prepared from mice lacking the small GTPase Rab3A, or, expressing a point mutation of Rab3A, Rab3A Earlybird, have an altered homeostatic plasticity response to a loss of network activity. The Rab3A Earlybird mutation was discovered in a mutagenesis screen in mice for genes involved in the rest-activity cycle (Kapfhamer et al., 2002 [↗](#)). The Earlybird mutant was named for its phenotype, a significantly shorter circadian period which resulted in early awakening. In that study, the authors concluded that the Earlybird mutation was likely dominant-negative, as the shift in circadian period was substantially more dramatic than that observed in the Rab3A^{-/-} mice. As described in the Introduction, the effects of loss of Rab3A have been modest, so we included experiments using the Rab3A Earlybird mutant mice to see if there would be a more robust phenotype, and in doing so, strengthen conclusions based on the Rab3A deletion mice. To obtain Rab3A^{-/-} and Rab3A^{Ebd/Ebd} homozygotes, we established two mouse colonies of heterozygous breeders with cultures prepared from pups derived from a final breeding pair of homozygotes. Although we backcrossed Rab3A^{+/-} with Rab3A^{+/-Ebd} for 11 generations, clear differences in mEPSC amplitudes in untreated cultures (see below) and in calcium current amplitudes in adrenal chromaffin cells (unpublished obs.) remained. Therefore, throughout this study we keep the two strains separate and there are two Rab3A^{+/+} or ‘wild type’ phenotypes.

We found that loss of Rab3A resulted in a greatly impaired homeostatic increase after activity blockade. Example current traces of spontaneously occurring mEPSCs recorded from pyramidal neurons in untreated (CON) 13-14 DIV cortical cultures and sister cultures treated with 500 nM TTX for 48 hr prepared from wild type and Rab3A^{-/-} mice in the Rab3A^{+/-} colony are shown in **Figures 1A and B** [↗](#), respectively. Average mEPSC waveforms from the same recordings are shown in **Figures 1C and D** [↗](#). The mean mEPSC amplitudes for 30 control and 23 TTX-treated neurons from Rab3A^{+/+} cultures are displayed in the box and whisker plot in **Figure 1E** [↗](#); after

activity blockade the average mEPSC amplitude increased from 13.9 ± 0.7 pA to 18.2 ± 0.9 pA. In contrast, in cultures prepared from Rab3A^{-/-} mice, the average mEPSC amplitude was not significantly increased, for 25 untreated cells and 26 TTX-treated cells (**Figure 1F**, 13.6 ± 0.1 pA vs. 14.3 ± 0.6 pA). We found that TTX treatment also resulted in an increase in mEPSC frequency in cultures prepared from Rab3A^{+/+} mice, as shown in **Figure 1G** (CON, 2.26 ± 0.37 sec⁻¹; TTX, 4.62 ± 0.74 sec⁻¹). This TTX-induced frequency increase was strongly reduced in cultures from Rab3A^{-/-} mice, but there were a few outliers with very high frequencies still present after TTX treatment (**Figure 1H**, CON, 2.74 ± 0.49 Hz; TTX, 3.23 ± 0.93 Hz).

We next examined homeostatic plasticity in cultures prepared from wild type mice in the Rab3A^{+Ebd} colony. Example current traces of mEPSCs, and average mEPSC waveforms, are shown in **Figures 2A and C**, respectively, for cortical cultures prepared from wild type mice in the Rab3A^{+Ebd} colony. Treatment with TTX for 48 hr led to a significant increase in the average mEPSC amplitude of 23 TTX-treated cells when compared to 20 untreated cells (**Figure 2E**, CON, 11.0 ± 0.6 pA; TTX, 15.0 ± 1.3 pA). We note here that while the two wild type strains respond very similarly to activity blockade, the mean mEPSC amplitude in untreated cultures was significantly different in the two wild type strains, 13.9 ± 0.7 pA, wild type from Rab3A^{+/+} colony; 11.0 ± 0.6 pA, wild type from Rab3A^{+Ebd} colony ($p = 0.004$, Kruskal-Wallis test).

We found a complete disruption of homeostatic plasticity in cortical cultures prepared from Rab3A^{Ebd/Ebd} mice, as can be seen in viewing example mEPSC traces and average mEPSC waveforms (**Figures 2B and D**, respectively). The lack of TTX effect was confirmed in a comparison of mEPSC amplitude means for 21 untreated and 22 TTX-treated cells (**Figure 2F**, CON, 15.1 ± 1.0 pA vs. TTX, 14.6 ± 1.1 pA). There was a trend towards increased mEPSC frequency in TTX-treated cultures from this wild type strain, but the difference was not as robust as for the wild type cultures from Rab3A^{+/+} colony, and did not reach statistical significance (**Figure 2G**, CON, 1.15 ± 0.19 sec⁻¹; TTX, 2.54 ± 0.55 sec⁻¹). This trend was not observed in neurons from cultures prepared from Rab3A^{Ebd/Ebd} mice, likely due to an increase in frequency in untreated cells—frequencies remained high in TTX-treated cells (**Figure 2H**, CON, 1.71 ± 0.41 sec⁻¹; TTX, 3.05 ± 0.80 sec⁻¹).

Our results show that the homeostatic increase in mEPSC amplitude after activity blockade is disrupted in both Rab3A^{-/-} and the Rab3A^{Ebd/Ebd} cortical neurons, strongly supporting a crucial role of functioning Rab3A in this process. However, it is important to note that the disruption differs for Rab3A^{-/-} and Rab3A^{Ebd/Ebd}. In the Rab3A^{-/-} data set, mEPSCs from untreated cultures were indistinguishable from mEPSCs from Rab3A^{+/+} untreated cultures, demonstrating loss of Rab3A has no impact on basal activity mEPSC amplitudes, but the increase in mEPSC amplitudes after activity blockade was strongly diminished in Rab3A^{-/-} neurons. In the Rab3A^{Ebd/Ebd} data set, mEPSC amplitudes from untreated cultures were significantly larger than those of untreated cultures from wild type mice, as can be seen comparing the CON values in **Figures 2E and F** (CON, Rab3A^{+/+}, 11.0 ± 0.7 pA, vs. CON, Rab3A^{Ebd/Ebd}, 15.1 ± 1.0 pA, $p = 0.0027$, Kruskal-Wallis test). The increase in mEPSC amplitude in cultures from Rab3A^{Ebd/Ebd} mice is consistent with the increase in mEPSC amplitude we observed at the Rab3A^{Ebd/Ebd} NMJ (Wang et al., 2011). Similar increases in mEPSC amplitude at baseline, combined with a failure to increase further after activity blockade, were noted in cultures prepared from mECP2, AKAP, Homer1a, and Arc/Arg3.1 deletion mice (Shepherd et al., 2006; Hu et al., 2010; Xu and Pozzo-Miller, 2017; Sanderson et al., 2018) as well as in cultures treated with ubiquitin proteasome inhibitors (Jakawich et al., 2010a) or microRNA 186-5p (Silva et al., 2019), suggesting the possibility of a generalized phenomenon in which increased mEPSC amplitude is induced when homeostatic regulation is disrupted. Since we did not observe such an increase in mEPSC amplitude at baseline in cultures from Rab3A^{-/-} mice, it remains a possibility that the point mutant may bind to novel partners, causing activities that would not be either facilitated or inhibited by Rab3A. Still, it is a strong coincidence that the novel activity of the mutant Earlybird Rab3A would affect mEPSC amplitude, the same characteristic that is modulated by activity blockade in a Rab3A dependent manner.

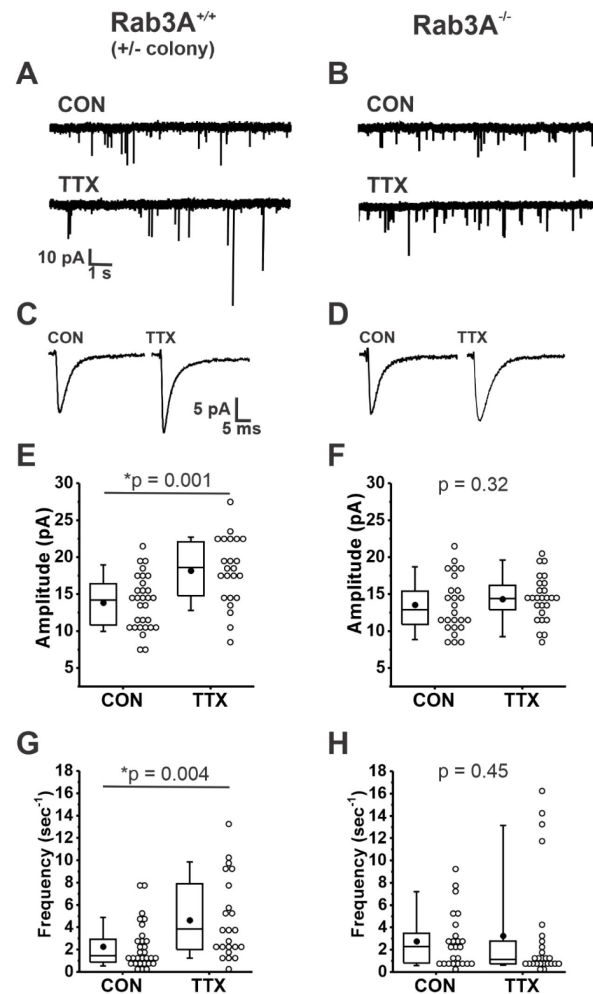


Figure 1.

Loss of Rab3A prevented the TTX-induced increases in amplitudes and frequency of mEPSCs recorded in cultured mouse cortical neurons.

(A) Ten second example traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from Rab3A^{+/+} mice from the Rab3A^{+/+} colony. (B) Ten second example traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from Rab3A^{-/-} mice. (C & D) Average traces for the recordings shown in A and B, respectively. (E) Box plots for average mEPSC amplitudes from untreated cells and TTX-treated cells in cultures prepared from Rab3A^{+/+} mice (Rab3A^{+/+} colony; CON, N = 30 cells, 13.9 ± 0.7 pA; TTX, N = 23 cells, 18.2 ± 0.9 pA; from 11 cultures). (F) Box plots for average mEPSC amplitudes from untreated cells and TTX-treated cells in cultures prepared from Rab3A^{-/-} mice (CON, N = 25 cells, 13.6 ± 0.7 pA; TTX, N = 26 cells, 14.3 ± 0.6 pA); from 11 cultures. (G) Box plots for average mEPSC frequency for same Rab3A^{+/+} cells as in (E) (CON, 2.26 ± 0.37 sec⁻¹; TTX, 4.62 ± 0.74 sec⁻¹). (H) Box plots for average mEPSC frequency for same Rab3A^{-/-} cells as in (F) (CON, 2.74 ± 0.49 sec⁻¹; TTX, 3.23 ± 0.93 sec⁻¹). Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test. For all p-values, * with underline indicates significance with p < 0.05.

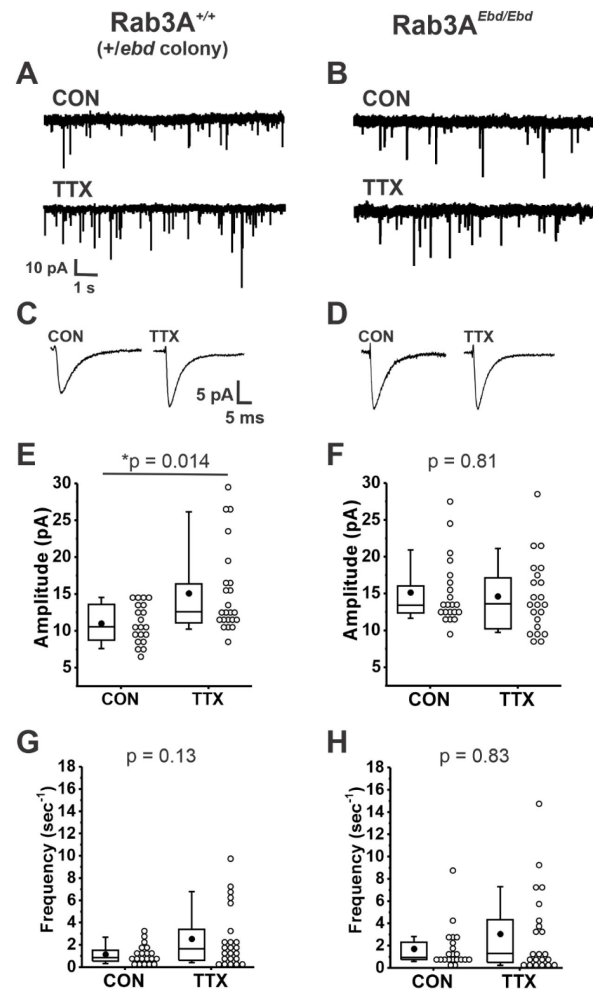


Figure 2.

Normally functioning Rab3A was required for TTX-induced homeostatic plasticity of mEPSC amplitudes in cultured mouse cortical neurons.

(A) Ten second example traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from *Rab3A*^{+/+} mice from the *Rab3A*^{+/Ebd} colony. (B) Ten second example traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from *Rab3A*^{Ebd/Ebd} mice. (C & D) Average traces for the recordings shown in A and B, respectively. (E) Box plots for average mEPSC amplitudes from untreated cells and TTX-treated cells from cultures prepared from *Rab3A*^{+/+} mice (*Rab3A*^{+/Ebd} colony; CON, N = 20 cells, 11.0 ± 0.6 pA; TTX, N = 23 cells, 15.1 ± 1.2 pA; from 6 cultures). (F) Box plots for average mEPSC amplitudes from untreated cells and TTX-treated cells from cultures prepared from *Rab3A*^{Ebd/Ebd} mice (CON, N = 21 cells, 15.1 ± 1.0 pA; TTX, N = 22 cells, 14.6 ± 1.1 pA; from 7 cultures.) (G) Box plots for average mEPSC frequency for same *Rab3A*^{+/+} cells as in (E) (CON, 1.15 ± 0.19 sec⁻¹; TTX, 2.54 ± 0.55 sec⁻¹). (H) Box plots for average mEPSC frequency for same *Rab3A*^{Ebd/Ebd} cells as in (F) (CON, 1.71 ± 0.41 sec⁻¹; TTX, 3.05 ± 0.80 sec⁻¹). Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test. For all p-values, * with underline indicates significance with p < 0.05.

Finally, although we cannot rule out that once mEPSC amplitude is increased at baseline, it is not possible for homeostatic plasticity to increase mEPSC amplitude further due to a non-specific ceiling effect, the simplest interpretation is that the presence of the mutant protein mimics the condition of activity blockade.

The small GTPase Rab3A is generally thought to function presynaptically to regulate synaptic vesicle trafficking, possibly in an activity-dependent manner (Castillo et al., 1997 [↗](#); Lonart et al., 1998 [↗](#); Leenders et al., 2001 [↗](#); Schluter et al., 2006 [↗](#); Coleman and Bykhovskaia, 2009 [↗](#); Tian et al., 2012 [↗](#)). In contrast, homeostatic plasticity of mEPSC amplitude has been attributed to an increase in postsynaptic receptors on the surface of the dendrite (O'Brien et al., 1998 [↗](#); Turrigiano et al., 1998 [↗](#)). Is Rab3A required for the increase in surface AMPA-type glutamate receptors that has been confirmed by multiple studies (see Introduction)? At the NMJ in vivo we could find no evidence for an increase in AChRs after TTX block of the sciatic nerve in vivo (Wang et al., 2005 [↗](#)). The type of AMPA receptor that has been shown to be increased, calcium-permeable GluA1-containing AMPA receptor or calcium-impermeable GluA2-containing AMPA receptor, appears to depend on the experimental manipulation. For example, block of activity by APV (NMDA antagonist) combined with TTX (inhibitor of Na⁺ channels) induces an increase in mEPSC amplitude in dissociated hippocampal cultures that is completely reversed by acute application of the Ca²⁺-permeable receptor-specific inhibitor NASPM (Sutton et al., 2006 [↗](#)), indicating the entirety of the homeostatic effect is due to an increase in homomeric Ca²⁺-permeable AMPA receptors, likely GluA1, although GluA3 and GluA4 are also Ca²⁺-permeable (Hollmann et al., 1991 [↗](#); Burnashev et al., 1992 [↗](#)). Heteromeric receptors of any of these subunits in combination with GluA2 would have rendered the receptors Ca²⁺-impermeable and no longer sensitive to NASPM (Blaschke et al., 1993 [↗](#); Bochet et al., 1994 [↗](#); Jonas and Burnashev, 1995 [↗](#)). In contrast, the increase in mEPSC amplitude in mouse hippocampal slice cultures induced by TTX alone was not affected by another Ca²⁺-permeable AMPA receptor inhibitor, philanthotoxin, suggesting mediation by either GluA2-containing AMPA receptors or a presynaptic effect on quantal size (Soden and Chen, 2010 [↗](#)); see also, (Dubes et al., 2022 [↗](#)).

Because we used TTX alone to block network activity, we expected that NASPM would not reverse the TTX-induced increase in mEPSC amplitude in our mouse cortical cultures, and we found that this was indeed the case. **Figure 3A** [↗](#) shows that the TTX-induced increase in mean mEPSC amplitude was nearly identical in a set of 11 CON and 11 TTX-treated cells before and after NASPM treatment (before NASPM, CON 12.9 ± 3.5 pA; TTX, 17.5 ± 3.1 pA; after NASPM, CON 11.9 ± 2.6 pA; TTX 16.1 ± 3.5 pA). This result indicates that the TTX-induced increase in mEPSC amplitude does not depend on Ca²⁺-permeable receptors, since the effect of TTX clearly remained when their presence was removed by acute NASPM application. We do not think that there was any technical issue with the NASPM application, because overall, mEPSC amplitudes were reduced modestly in both untreated and TTX-treated cultures (**Figure 3B** [↗](#), CON, before NASPM, 12.9 ± 3.5 pA; after, 11.9 ± 2.6 pA; TTX, before NASPM, 17.5 ± 3.1 pA; after, 16.1 ± 3.5 pA). Furthermore, NASPM consistently reduced mEPSC frequency (**Figure 3C** [↗](#), CON, before NASPM, 1.84 ± 0.55 sec⁻¹; after NASPM, 1.56 ± 0.53 sec⁻¹; TTX, before NASPM, 4.40 ± 3.51 sec⁻¹; after NASPM, 2.68 ± 2.25 sec⁻¹). Similar to the data in **Figure 1** [↗](#), frequency is significantly increased after TTX treatment (p = 0.04), but this difference is no longer significant after NASPM application (p = 0.20). A loss in frequency after acute application of NASPM suggests there are synaptic sites that express only homomers of Ca²⁺-permeable receptors; mEPSCs from these sites would be expected to be completely blocked by NASPM (see cartoon description in **Figure 3D** [↗](#), Left). The magnitude of the frequency reduction following acute NASPM appears to be greater after TTX treatment, suggesting that loss of activity promotes the establishment of synaptic sites that contain only Ca²⁺-permeable receptors. However, these new sites do not appear to contribute to the increase in mEPSC amplitude (**Figure 3A** [↗](#)). The lack of effect of NASPM also rules out the contribution of Ca²⁺-permeable forms of Kainate receptors GluA5 and GluA6 (Koike et al., 1997 [↗](#); Sun et al., 2009 [↗](#)).

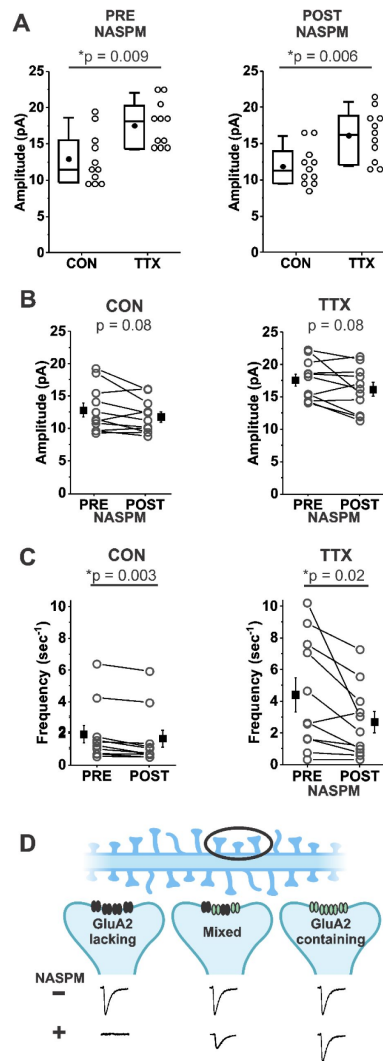


Figure 3

Figure 3.

Homeostatic plasticity of mEPSC amplitudes in mouse cortical cultures treated with TTX for 48 hr was not affected by acute inhibition of Ca²⁺ permeable AMPA receptors by NASPM (20 μ m).

(A) Box plot comparison of the TTX effect on mEPSC amplitudes in the same pyramidal neurons before and after application of 20 μ M NASPM. (N = 11 cells from 3 cultures; Pre-NASPM, CON: 12.9 ± 1.1 pA; TTX: 17.5 ± 0.9 pA; post-NASPM, CON: 11.9 ± 0.8 pA; TTX: 16.1 ± 1.0 pA). (B) Line series plot of mEPSC amplitudes before and after acute perfusion with 20 μ M NASPM for untreated and TTX-treated pyramidal neurons; same cells as in (A); CON, pre-NASPM: 12.9 ± 1.1 pA; post-NASPM: 11.9 ± 0.8 pA; TTX, pre-NASPM: 17.5 ± 0.9 pA; post-NASPM: 16.1 ± 0.1 pA. (C) Line series plot of mEPSC frequency before and after acute perfusion with 20 μ M NASPM for untreated and TTX-treated pyramidal neurons; same cells as in (A); CON, pre-NASPM: 1.84 ± 0.55 sec⁻¹; post-NASPM: 1.56 ± 0.53 sec⁻¹; TTX, pre-NASPM: 4.40 ± 1.06 sec⁻¹; post-NASPM: 2.68 ± 0.68 sec⁻¹. (D) A proposed mechanism for why NASPM had a robust effect on frequency without greatly affecting amplitude. A dendrite with spines (Top) is expanded on three postsynaptic sites (Middle) to show possible types of AMPA receptor distributions: Left, at a site comprised only of Ca²⁺-permeable AMPA receptors (NASPM-sensitive, GluA2-lacking receptors (black)), NASPM would completely inhibit the mEPSC, causing a decrease in frequency to be measured in the overall population; Right, at a site comprised only of Ca²⁺-impermeable AMPA receptors (NASPM-insensitive, GluA2-containing receptors (green)), NASPM would have no effect on mEPSC amplitude; Middle, at a site comprised of a mix of Ca²⁺-permeable and Ca²⁺-impermeable AMPA receptors, NASPM would partially inhibit the mEPSC. Diagram produced in BioRender.com (2024). Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test. Line series plot p values are from student's paired t test. For all p-values, * with underline indicates significance with $p < 0.05$.

Having established that homomeric Ca^{2+} -permeable receptors are not contributing to the homeostatic increase in mEPSC amplitude, we turned to immunohistochemistry and confocal imaging to assess whether GluA2 receptor expression, which will identify either GluA2 homomers or GluA1-GluA2 heteromers, was increased in our wild type mouse cortical cultures following 48 hr treatment with TTX. Since mEPSCs necessarily report synaptic levels of receptors, we used VGLUT1-immunofluorescence to identify synapses on pyramidal primary apical dendrites labeled with MAP-2 immunofluorescence. We focused on the primary dendrite of pyramidal neurons as a way to reduce variability that might arise from being at widely ranging distances from the cell body, or, from inadvertently sampling dendritic regions arising from other neurons, possibly inhibitory neurons. **Figure 4** (Top) shows two pyramidal neurons, one from an untreated culture on the left, and one from a TTX-treated culture on the right, prepared from Rab3A^{+/+} mice. To measure characteristics of synaptically located GluA2 receptor clusters, we zoomed in on the primary dendrite (region in white rectangle). The zoomed regions for single confocal sections of the selected area are shown below, and contain pairs of VGLUT1- and GluA2-immunofluorescent clusters indicated by white trapezoidal frames. A dendrite typically required ~10 confocal sections to be fully captured, and the total number of synaptic pairs for all the sections imaged on a dendrite was usually < 20, so this is an atypically high number of pairs within a single section; these particular dendrites and sections were selected for illustration purposes. In addition to the synaptic pairs, we observed many GluA2-immunofluorescent clusters not associated with VGLUT1 immunofluorescence (a few are indicated with white arrows), most probably extrasynaptic receptors. There were also GluA2-positive clusters present outside of MAP2-positive dendrites, which may be located on astrocytes (Fan et al., 1999). Finally, GluA2 immunofluorescent clusters close to VGLUT1 immunofluorescence but not located along any apparent MAP2-positive neurites suggests the presence of axon-axonal contacts, although VGLUT1 has also been detected in astrocytes (Ormel et al., 2012). Only sites that contained both VGLUT1 and GluA2 immunofluorescence close to the primary MAP2-positive dendrite or a secondary branch along this primary dendrite were selected for analyses. In the experiment from which these images were collected, we analyzed the distance from the cell body for each synapse. The average distance from the cell body, for dendrites from the untreated cell, was $38.5 \pm 2.8 \mu\text{m}$; for the TTX-treated cell, it was $42.4 \pm 3.2 \mu\text{m}$ ($p = 0.35$, Kruskal-Wallis test). Since the greatest distance from the cell body is set by the outer limit of the zoom window, and that window was placed adjacent to the cell body, distances in other experiments would be in this same range.

Variability in the magnitude of the homeostatic response from culture to culture is averaged out in physiological experiments by the pooling of data from many cells across many cultures. To reduce the necessity for many cultures, we chose to pair experiments in the same cultures by recording mEPSCs from one set of coverslips, and processing another set of coverslips from the same culture for immunofluorescence. We completed this matched paradigm of physiology and immunofluorescence, which to our knowledge has never before been done, for 3 cultures prepared from Rab3A^{+/+} mice and 3 cultures prepared from Rab3A^{-/-} mice. We present the results for mEPSC data and imaging data pooled across the 3 experiments in **Figure 5**, with the data summarized in **Table 1**. Levels of GluA2 immunofluorescence at synaptic sites were quantified by the size of the GluA2-immunofluorescent receptor cluster, the average intensity value, and the integral of intensity values across the area of the cluster. In the data pooled from the 3 matched experiments, neurons from Rab3A^{+/+} cultures showed a significant increase in mean mEPSC amplitudes following activity blockade (**Figure 5A**; **Table 1**). This result indicates that the homeostatic response averaged across the 3 cultures was very similar to, but slightly smaller than, that of the data set presented in **Figure 1**. The means for size, intensity and integral of GluA2 immunofluorescent receptor clusters showed trends to higher values after activity blockade, but the differences did not reach statistical significance (**Figure 5B** size; **Figure 5C** intensity; **Figure 5D** integral; **Table 1**). These results support previous studies demonstrating an increase in AMPA receptors following activity blockade, with the magnitude of the changes in GluA2 receptor cluster size and integral matching that of mEPSC amplitude, at around 20% (**Table 1**), but not achieving the same level of statistical significance.

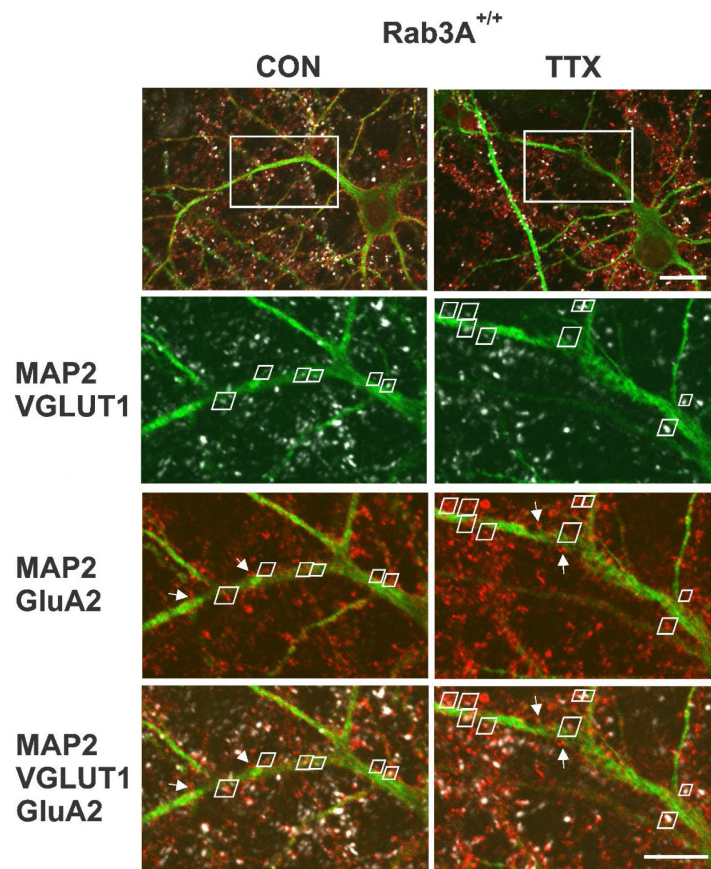


Figure 4.

Identification of synaptic GluA2 receptor immunofluorescence on primary dendrites of pyramidal neurons in high density mouse cortical co-cultures prepared from Rab3A^{+/+} mice.

(Top) Non-zoomed single confocal sections collected with a 60X oil immersion objective of recognizably pyramidal-shaped neurons, presumed to be excitatory neurons, selected for synaptic GluA2 analysis. A neuron was selected from an untreated coverslip (CON, Left) and a TTX-treated coverslip (TTX, Right) from the same culture prepared from Rab3A^{+/+} animals (= Rab3A^{+/+} Culture #2 in [Figure 6](#)). The white rectangular boxes indicate 5X zoomed areas shown in the images below. Note that the depth of the single confocal section of the non-zoomed neuron image is not at the same depth as the confocal section in zoomed dendritic image, so some features are not visible in both images. Scale bar, 20 μ m. (Bottom) 5X zoom single confocal sections selected for demonstration purposes because they had an unusually high number of identified synaptic pairs along the primary dendrite contained within a single confocal section. Synaptic pairs, highlighted with white trapezoids, were identified based on close proximity of GluA2 (red) and VGLUT1 (white) immunofluorescence, apposed to the MAP2 immunofluorescent primary dendrite (green). Some apparent synaptic pairs are not highlighted with a white trapezoid because there was a different confocal section in which the two immunofluorescent sites were maximally bright. There were GluA2 positive clusters located on the primary dendrites that were not apposed to VGLUT1-positive terminals (4 of these non-synaptic GluA2 clusters are highlighted with white arrows in the MAP2-GluA2 and MAP2-GluA2-VGLUT1 panels). Scale bar, 10 μ m.

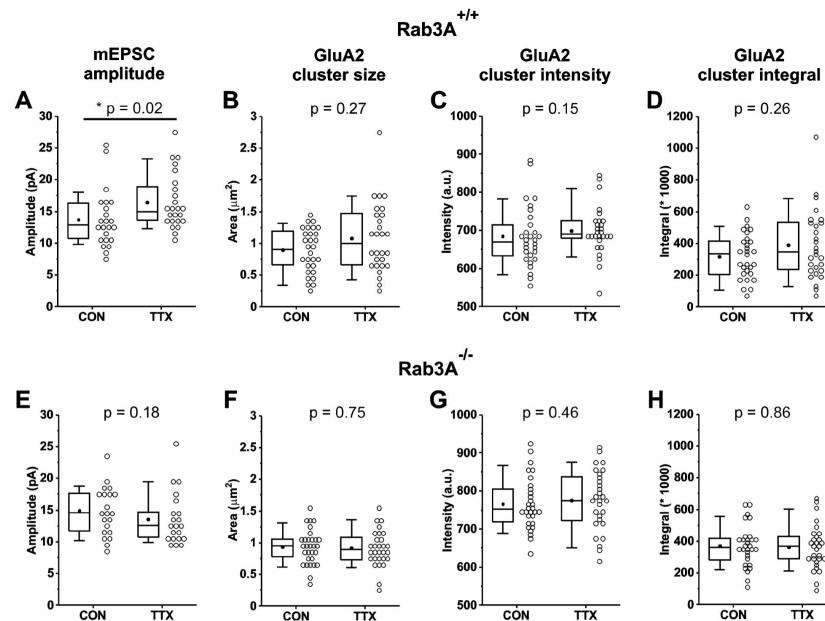


Figure 5.

Comparison of mEPSC amplitude and GluA2 receptor cluster data, pooled from 3 cortical co-cultures prepared from Rab3A^{+/+} mice and 3 co-cultures prepared from Rab3A^{-/-} mice.

(A) Mean mEPSC amplitudes recorded in pyramidal neurons from untreated (CON) and TTX-treated coverslips (TTX) of co-cultures prepared from Rab3A^{+/+} mice (CON, N = 23 cells, 13.7 ± 0.9 pA; TTX, N = 24 cells, 16.4 ± 0.9 pA; from 3 cultures). (B) Mean sizes of synaptic GluA2 receptor clusters, identified by proximity to VGLUT1-positive terminals, on primary dendrites of pyramidal neurons of untreated and TTX-treated coverslips from the same cultures as in (A) (CON, N = 29 dendrites, $0.90 \pm 0.06 \mu\text{m}^2$; TTX, N = 28 dendrites, $1.08 \pm 0.10 \mu\text{m}^2$). (C) Mean intensities of the same GluA2 receptor clusters as in (B) (CON, 684 ± 15 A.U.; TTX, 699 ± 12 A.U.). (D) Mean integrals of the same GluA2 receptor clusters as in (B). (CON, $317,898 \pm 26,726$ A.U.; TTX, $389,487 \pm 41,465$ A.U.). (E) Mean mEPSC amplitudes in pyramidal neurons from untreated and TTX-treated coverslips of mouse cortical co-cultures from Rab3A^{-/-} mice (CON, N = 21 cells, 14.9 ± 0.8 pA; TTX, N = 21 cells, 13.5 ± 0.9 pA; from 3 cultures). (F) Mean sizes of GluA2 receptor clusters, identified by proximity to VGLUT1-positive terminals, on primary dendrites of pyramidal neurons of untreated and TTX-treated coverslips from the same cultures as in (E). (CON, N = 30 dendrites, $0.93 \pm 0.05 \mu\text{m}^2$; TTX, N = 30 dendrites, $0.92 \pm 0.05 \mu\text{m}^2$). (G) Mean intensities of the same GluA2 receptor clusters as in (F). (CON, 766 ± 12 A.U.; TTX, 776 ± 78 A.U.). (H) Mean integrals of the same GluA2 receptor clusters as in (F). (CON, $369,436 \pm 23,439$ A.U.; TTX, $366,389 \pm 25,049$ A.U.). Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10 and 90 percentiles; open circles indicate means of individual cells; line, median; dot, mean. p values (shown on the graphs) were determined with Kruskal-Wallis test. For all p-values, * with underline indicates significance with $p < 0.05$.

Table 1.

Comparison of mEPSC amplitude and GluA2 receptor cluster characteristics in the same mouse cortical cultures prepared from Rab3A^{+/+} mice, and another set of cultures prepared from Rab3A^{-/-} mice.

Data pooled from 3 cultures, and data from each of the 3 individual cultures.

	Rab3A ^{+/+}				Rab3A ^{-/-}			
	Pooled Data from 3 cultures				Pooled Data from 3 cultures			
Quantity	CON Mean, SE (N)	TTX Mean, SE (N)	% change	p value	CON Mean, SE (N)	TTX Mean, SE (N)	% change	p value
mEPSC amplitude	13.7 ± 0.9 pA (23)	16.4 ± 0.9 pA (24)	+19.7	*p = 0.02	14.9 ± 0.8 pA (21)	13.5 ± 0.9 pA (21)	-9.4	p = 0.18
GluA2 cluster size	0.90 ± 0.06 μm ² (29)	1.08 ± 0.10 μm ² (28)	+20.0	p = 0.27	0.93 ± 0.05 μm ² (30)	0.91 ± 0.05 μm ² (29)	-2.2	p = 0.75
GluA2 cluster intensity	684 ± 15 a.u. (29)	699 ± 12 a.u. (28)	+2.2	p = 0.15	766 ± 12 a.u. (30)	776 ± 15 a.u. (29)	+1.3	p = 0.46
GluA2 cluster integral	317898 ± 26726 a.u. (29)	389487 ± 41465 a.u. (28)	+22.5	p = 0.26	369436 ± 23439 a.u. (30)	364237 ± 25833 a.u. (29)	-0.8	p = 0.86
Quantity	Rab3A ^{+/+} Culture #1				Rab3A ^{-/-} Culture #1 (not same culture as Rab3A ^{+/+})			
mEPSC amplitude	14.2 ± 2.4 pA (6)	15.9 ± 1.9 pA (7)	+12.0	p = 0.25	15.4 ± 1.2 pA (7)	14.4 ± 0.8 pA (6)	-6.5	p = 0.20
GluA2 cluster size	0.73 ± 0.09 μm ² (10)	0.89 ± 0.09 μm ² (10)	+21.9	p = 0.45	0.87 ± 0.07 μm ² (10)	0.80 ± 0.12 μm ² (10)	-8.0	p = 0.41
GluA2 cluster intensity	682 ± 15 a.u. (10)	704 ± 11 a.u. (10)	+3.2	p = 0.13	735 ± 10 a.u. (10)	731 ± 25 a.u. (10)	-0.5	p = 0.71
GluA2 cluster integral	256364 ± 35242 a.u. (10)	316647 ± 35077 a.u. (10)	+23.5	p = 0.29	328477 ± 28250 a.u. (10)	302594 ± 53052 a.u. (10)	-7.9	p = 0.41
Quantity	Rab3A ^{+/+} Culture #2				Rab3A ^{-/-} Culture #2 (not the same culture as Rab3A ^{+/+})			
mEPSC amplitude	13.8 ± 2.2 pA (7)	17.7 ± 1.8 pA (8)	+28.2	p = 0.16	15.2 ± 1.8 pA (8)	13.5 ± 1.8 pA (9)	-11.2	p = 0.50
GluA2 cluster size	0.91 ± 0.12 μm ² (9)	1.42 ± 0.24 μm ² (9)	+56.0	*p = 0.05	1.05 ± 0.10 μm ² (10)	0.91 ± 0.04 μm ² (9)	-13.3	p = 0.37
GluA2 cluster intensity	621 ± 14 a.u. (9)	679 ± 26 a.u. (9)	+9.3	*p = 0.04	800 ± 26 a.u. (10)	782 ± 23 a.u. (9)	-2.2	p = 0.51
GluA2 cluster integral	296300 ± 44108 a.u. (9)	512336 ± 96892 a.u. (9)	+72.9	*p = 0.04	425552 ± 49486 a.u. (10)	354289 ± 23099 a.u. (9)	-16.7	p = 0.33
Quantity	Rab3A ^{+/+} Culture #3				Rab3A ^{-/-} Culture #3 (not the same culture as Rab3A ^{+/+})			
mEPSC amplitude	13.4 ± 0.8 pA (10)	15.7 ± 1.0 pA (9)	+17.2	p = 0.10	13.9 ± 1.2 pA (6)	12.9 ± 1.5 pA (6)	-7.2	p = 0.26
GluA2 cluster size	1.05 ± 0.11 μm ² (10)	0.95 ± 0.15 μm ² (9)	-9.5	p = 0.46	0.89 ± 0.08 μm ² (10)	1.03 ± 0.08 μm ² (10)	+15.7	p = 0.29
GluA2 cluster intensity	744 ± 28 a.u. (10)	714 ± 25 A.U. (9)	-4.0	p = 0.29	765 ± 22 a.u. (10)	814 ± 23 a.u. (10)	+6.4	p = 0.17
GluA2 cluster integral	398870 ± 49500 a.u. (10)	347571 ± 65454 a.u. (9)	-12.9	p = 0.46	354279 ± 38755 a.u. (10)	434833 ± 42350 a.u. (10)	+22.7	p = 0.29

As expected, mean mEPSC amplitude was not increased following activity blockade in the data pooled from the new set of 3 Rab3A^{-/-} cultures (**Figure 5E** [↗](#)). For immunofluorescence results from the same Rab3A^{-/-} cultures, no trend towards higher values was apparent for GluA2 receptor cluster characteristics (**Figure 5F** [↗](#) size; **Figure 5G** [↗](#) intensity; **Figure 5H** [↗](#) integral; **Table 1** [↗](#)). Taken together, these results indicate that the modest increases in GluA2 receptor cluster size and integral following activity blockade observed in wild-type cultures do not occur in the absence of Rab3A. These results suggest the surprising finding that the presynaptic vesicle protein Rab3A is involved in the regulation of postsynaptic AMPA receptors during homeostatic plasticity.

As noted above, the homeostatic increase in mEPSC amplitude in cultures from Rab3A^{+/+} mice reached significance whereas that of receptor levels did not. To determine where this disparity was coming from, we compared mEPSC amplitude and GluA2 receptor data within individual experiments. In order to evaluate whether a particular culture displayed homeostatic plasticity of mEPSC amplitude, we set a minimum requirement of recording from 6 cells per condition, meaning a successful experiment required at least 12 usable recordings in one day (6 CON and 6 TTX; i.e. low holding current, stable baseline, low noise, etc.). Ultimately, our sample size ranged from 6 to 10 cells per condition per culture. **Figures 6A-C** [↗](#) show the box plots for mEPSC amplitudes, and GluA2 receptor cluster size, intensity and integrals, for 3 individual Rab3A^{+/+} cultures. With these small sample sizes, the differences between untreated and TTX-treated cells were not statistically significant for mEPSC amplitudes in any of the 3 cultures, but an upward trend occurred in every case, ranging from 12 to 28% (**Table 1** [↗](#)). Second, in two of the three Rab3A^{+/+} experiments (Cultures #1 and #2) GluA2 receptor cluster size increased 22 and 56%, respectively, significantly so for Culture #2 (**Table 1** [↗](#)). However, in one experiment (Culture #3), GluA2 receptor cluster size showed a downward trend of 9.5%. Because we have 10 CON mean mEPSC amplitudes and 9 TTX mean mEPSC amplitudes that indicate this particular culture very likely has a normal homeostatic increase in mEPSC amplitude induced by activity blockade, the lack of increase in GluA2 receptor cluster characteristics suggests that mEPSC amplitudes can increase in the absence of an increase in receptors. Furthermore, Culture #3's opposite results from Cultures #1 and #2, explain why in the pooled data set, the increase in GluA2 receptor characteristics after TTX treatment do not reach statistical significance.

Figures 7A-C [↗](#) show the box plots for mEPSC amplitudes, and GluA2 receptor cluster size, intensity and integrals, for 3 individual Rab3A^{-/-} cultures. Here, mEPSC amplitudes showed a downward trend in all 3 cultures, as did GluA2 receptor characteristics for two of the three cultures. But we found another obvious mismatch between the mEPSC amplitude data and the GluA2 receptor cluster data for one of the Rab3A^{-/-} cultures. For Culture #3 (**Figure 7C** [↗](#)), the box plot of mEPSC mean amplitudes showed a downward trend after TTX treatment, decreasing by 7% (**Table 1** [↗](#)), whereas GluA2 receptor cluster size, intensity and integral all showed upward trends, the increase ranging from 6 to 23% (**Table 1** [↗](#)). This result may suggest that the disruption of the homeostatic increase in receptors in the absence of Rab3A is not absolute. It also indicates that synaptic GluA2 receptor levels can increase without an accompanying increase in mEPSC amplitudes.

In summary, our matched mEPSC and receptor cluster results indicate that: 1. activity blockade resulted in an increase in mEPSC amplitudes usually accompanied by an increase in synaptic GluA2 receptors; 2. loss of Rab3A disrupted both the increase in mEPSC amplitudes and the increase in GluA2 receptor levels; and 3. The differences between homeostatic responses of mEPSC amplitudes and the GluA2 cluster sizes within the same cultures suggest there may be other factors that contribute to mEPSC amplitude.

Having used VGLUT1 immunofluorescence to mark synaptic sites, we could also ask whether there was a presynaptic effect of activity blockade on the size, intensity or integral of the VGLUT1 signal at sites apposed to the previously examined GluA2 postsynaptic sites. In the pooled data from cultures prepared from Rab3A^{+/+} mice, we found no change in the size of the VGLUT1-positive

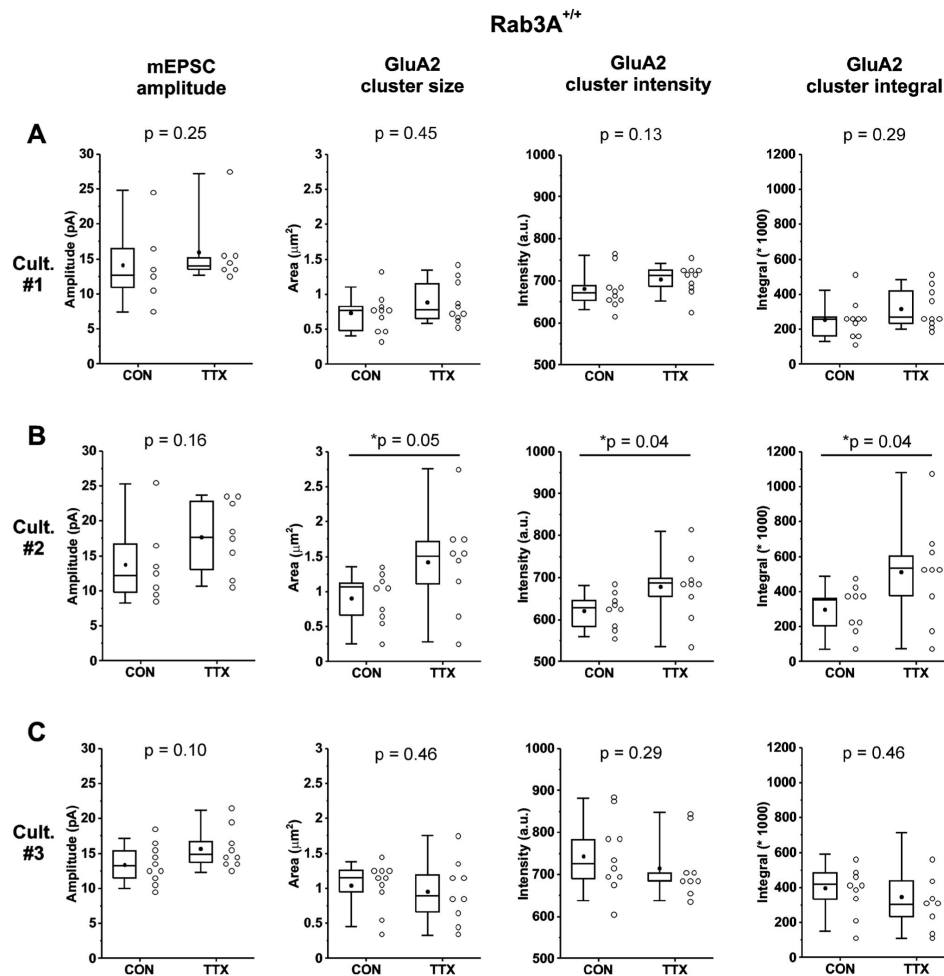


Figure 6.

In cultures prepared from Rab3A^{+/+} mice, increases in mEPSC amplitudes following TTX treatment were not always matched by increases in GluA2 receptor cluster characteristics.

(A) Culture #1: mean mEPSC amplitudes for pyramidal neurons from untreated coverslips (CON) and TTX-treated coverslips (TTX) (CON, N = 6, 14.2 ± 2.2 pA; TTX, N = 6, 15.9 ± 1.9 pA), compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken of dendrites from other coverslips in the same culture (CON, N = 10, TTX, N = 9; size, CON, 0.73 ± 0.09 μm^2 ; TTX, 0.89 ± 0.09 μm^2 ; intensity, CON, 682 ± 15 A.U.; TTX, 704 ± 11 A.U.; integral, CON, $256,364 \pm 35,242$ A.U.; TTX, $316,647 \pm 35,077$ A.U.). (B) Culture #2: mean mEPSC amplitudes for pyramidal neurons from untreated and TTX-treated coverslips (CON, N = 7, 13.8 ± 2.4 pA; TTX, N = 8, 17.7 ± 1.8 pA), compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken of dendrites from other coverslips from the same culture (CON, N = 9, TTX, N = 9; size, CON, 0.91 ± 0.12 μm^2 ; TTX, 1.42 ± 0.24 μm^2 ; intensity, CON, 621 ± 14 A.U.; TTX, 679 ± 26 A.U.; integral, CON, $296,300 \pm 44,108$ A.U.; TTX, $512,336 \pm 96,892$ A.U.). (C) Culture #3: mean mEPSC amplitudes for pyramidal neurons from untreated coverslips and TTX-treated coverslips (CON, N = 10, 13.4 ± 0.8 pA; TTX, N = 9, 15.7 ± 1.0 pA), compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken from other coverslips from the same culture. (CON, N = 10, TTX, N = 9; size, CON, 1.05 ± 0.11 μm^2 ; TTX, 0.95 ± 0.15 μm^2 ; intensity, CON, 744 ± 28 A.U.; TTX, 714 ± 25 A.U.; integral, CON, $398,870 \pm 49,500$ A.U.; TTX, $347,571 \pm 65,454$ A.U.). Cultures were fixed and processed for immunofluorescence on the same day as mEPSC recordings, and imaged within 1 week. Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test. p-values denoted with * and underline indicates significance of $p < 0.05$.

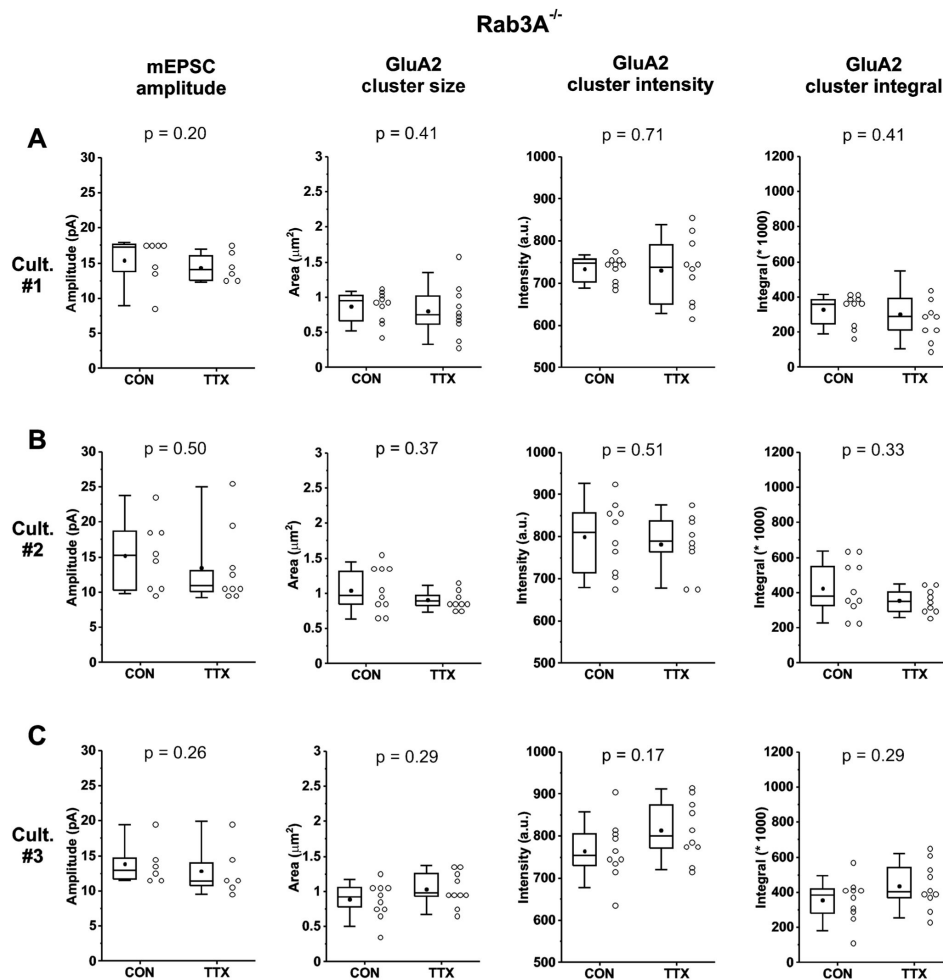


Figure 7.

In cultures prepared from Rab3A^{-/-} mice, a lack of effect of TTX treatment in mEPSC amplitudes was not always matched by a lack of effect in synaptic GluA2 receptor cluster characteristics.

(A) Culture #1: mean mEPSC amplitudes for pyramidal neurons from untreated coverslips (CON) and TTX-treated coverslips (TTX) (CON, N = 7, 15.4 ± 1.2 pA; TTX, N = 6, 14.4 ± 0.8 pA), compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken of dendrites from other coverslips in the same culture (CON, N = 10, TTX, N = 10; size, CON, $0.87 \pm 0.07 \mu\text{m}^2$; TTX, $0.80 \pm 0.12 \mu\text{m}^2$; intensity, CON, 735 ± 10 A.U.; TTX, 731 ± 25 A.U.; integral, CON, $328,477 \pm 28,250$; TTX, $302,594 \pm 53,052$ A.U.). (B) Culture #2: mean mEPSC amplitudes for pyramidal neurons from untreated and TTX-treated coverslips (CON, N = 8, 15.2 ± 1.8 pA; TTX, N = 9, 13.5 ± 1.8 pA), compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken of dendrites from other coverslips in the same culture (CON, N = 10, TTX, N = 10; size, CON, $1.05 \pm 0.10 \mu\text{m}^2$; TTX, $0.93 \pm 0.05 \mu\text{m}^2$; intensity, CON, 800 ± 26 A.U.; TTX, 781 ± 21 A.U.; integral, CON, $425,552 \pm 49,486$ A.U.; TTX, $361,740 \pm 21,963$ A.U.). (C) Culture #3: mean mEPSC amplitudes for pyramidal neurons from untreated and TTX-treated coverslips (CON, N = 6, 13.9 ± 1.2 pA; TTX, N = 6, 12.9 ± 1.5 pA) compared to mean sizes, intensities, and integrals of synaptic GluA2 receptor clusters in images taken of dendrites from other coverslips in the same culture (CON, N = 10, TTX, N = 10; size, CON, $0.89 \pm 0.08 \mu\text{m}^2$; TTX, $1.03 \pm 0.08 \mu\text{m}^2$; intensity, CON, 765 ± 22 A.U.; TTX, 814 ± 23 A.U.; integral, CON, $354,279 \pm 38,755$ A.U.; TTX, $434,833 \pm 42,350$ A.U.). Cultures were fixed and processed for immunofluorescence on the same day as mEPSC recordings, and imaged within 1 week. Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test.

regions, and trends towards reduced intensity and integral (**Table 2**). These data do not support the idea that activity blockade caused an increase in the amount of VGLUT1 per vesicle, although we cannot rule out that a simultaneous reduction in the number of vesicles obscured the effect, since we did not have an independent label for synaptic vesicles. In cultures prepared from Rab3A^{-/-} mice, the size, intensity and integral of the VGLUT1-positive regions showed upward trends (**Table 2**), suggesting the possibility that the modest decreases observed for cultures prepared from wild type mice depend on Rab3A, but it is difficult to make any conclusions when the p values are far from reaching significance. Similarly to the GluA2 receptor cluster data, the likely reason that the pooled data effects did not reach significance is because the effects were not consistent across all three cultures, with one Rab3A^{+/+} culture showing an upward trend instead of a downward trend, and one Rab3A^{-/-} culture showing a downward trend instead of an upward trend (**Table 2**). It should be noted that there was no consistent inverse pattern for GluA2 vs. VGLUT1. While in cultures prepared from Rab3A^{-/-} mice, when GluA2 characteristics trended downward, VGLUT1 characteristics trended upward, this pattern did not hold for cultures prepared from Rab3A^{+/+} mice. In Rab3A^{+/+} Culture #2, GluA2 showed significant increases, and VGLUT1 showed essentially no change. In Rab3A^{+/+} Culture #3, both GluA2 and VGLUT1 trended downwards. Taken together, the GluA2 and VGLUT1 immunofluorescence data indicate that GluA2 receptors likely contribute to the increase in mEPSC amplitudes after activity blockade in Rab3A^{+/+} cultures, and the disruption of that increase in Rab3A^{-/-} cultures, but VGLUT1 levels change in the opposite direction from mEPSC amplitudes. Interestingly, VGLUT1 levels are strongly affected by loss of Rab3A, with a highly significant decrease in integral in untreated cultures from Rab3A^{-/-} mice, compared to untreated cultures from Rab3A^{+/+} mice (size, $p = 0.019$; intensity, $p = 0.004$; integral, $p = 4.6 \times 10^{-4}$; **Table 2**).

We have demonstrated here for the first time that the synaptic vesicle protein Rab3A influences the homeostatic regulation of postsynaptic GluA2 receptors. What are possible ways that Rab3A could exert a postsynaptic action? It has previously been shown that exogenous addition of TNF α to hippocampal cultures causes an increase in surface expression of GluA1 receptors (although not GluA2 receptors), and that the homeostatic increase in mEPSC amplitudes is abolished in cultures prepared from the TNF α deletion mouse (Stellwagen et al., 2005; Stellwagen and Malenka, 2006). Furthermore, neurons from TNF α ^{+/-} mice plated on glial feeder layers derived from TNF α ^{-/-} mice fail to show the increase in mEPSC amplitude after TTX treatment, indicating that the TNF α inducing the receptor increases following activity blockade comes from the glial cells (Stellwagen and Malenka, 2006). It has recently been shown that the source of TNF α is astrocytes rather than microglia (Heir et al., 2024). Rab3A has been detected in astrocytes (Maienschein et al., 1999; Hong et al., 2016), so to determine whether Rab3A is acting via regulating TNF α release from astrocytes, we performed experiments similar to those of Stellwagen and Malenka (2006) (schema illustrated in **Figure 8**, Left). It should be noted that under our culturing conditions, the glial feeder layers would be expected to be composed mainly of astrocytes (Heir et al., 2024). We compared the effect of activity blockade on mEPSC amplitudes recorded from cortical neurons from Rab3A^{+/-} mice plated on Rab3A^{+/-} glial feeder layers (**Figure 8A**); neurons from Rab3A^{+/-} mice plated on Rab3A^{-/-} glia (**Figure 8B**), and neurons from Rab3A^{-/-} mice plated on Rab3A^{+/-} glia (**Figure 8C**). If Rab3A is required for TNF α release from glia, then Rab3A^{+/-} neurons plated on Rab3A^{-/-} glia should not show a homeostatic increase in mEPSC amplitude after treatment with TTX, and any cultures with Rab3A^{+/-} glia should have a normal homeostatic response. We found the opposite result: mEPSC amplitudes increased dramatically in cultures where Rab3A was present in neurons (WT on WT, CON, 13.3 ± 0.5 pA, TTX, 16.7 ± 1.2 pA; WT on KO, CON, 13.3 ± 1.0 pA, TTX, 18.8 ± 1.4 pA), but the increase was greatly diminished in cultures where Rab3A was present only in glia (KO on WT, CON, 15.2 ± 1.1 pA, TTX, 16.9 ± 0.7 pA). It is interesting to note that in the glial feeder layer culture, the loss of Rab3A in the neurons may have caused an increase in mEPSC amplitude at baseline, similar to the Earlybird mutant in dissociated cultures, but it is also possible this can be attributed to culture to culture variability.

Table 2.

Comparison of mEPSC amplitude and VGLUT1 positive presynaptic site characteristics in the same mouse cortical cultures prepared from Rab3A^{+/+} and Rab3A^{-/-} mice.

Data pooled from 3 cultures, and data from individual cultures. mEPSC data are identical to that in [Table 1](#), reproduced here for comparison purposes.

	Rab3A ^{+/+} Pooled Data from 3 cultures				Rab3A ^{-/-} Pooled Data from 3 cultures			
Quantity	CON Mean, SE (N)	TTX Mean, SE (N)	% change	p value	CON Mean, SE (N)	TTX Mean, SE (N)	% change	p value
mEPSC amplitude	13.7 ± 0.9 pA (23)	16.4 ± 0.9 pA (24)	+19.7	*p = 0.02	14.9 ± 0.8 pA (21)	13.5 ± 0.9 pA (21)	-9.4	p = 0.18
VGLUT1 site size	1.17 ± 0.08 μm ² (29)	1.07 ± 0.06 μm ² (24)	-7.7	p = 0.97	0.89 ± 0.03 μm ² (30)	0.95 ± 0.04 μm ² (29)	+6.7	p = 0.55
VGLUT1 site intensity	699 ± 39 a.u. (29)	639 ± 31 a.u. (28)	-8.6	p = 0.28	554 ± 13 a.u. (30)	570 ± 19 a.u. (29)	+2.9	p = 0.18
VGLUT1 site integral	430787 ± 36818 a.u. (29)	351653 ± 21689 a.u. (28)	-18.4	p = 0.13	257159 ± 18265 a.u. (30)	289857 ± 18521 a.u. (29)	+12.7	p = 0.21
Quantity	Rab3A ^{+/+} Culture #1				Rab3A ^{-/-} Culture #1 (not same culture as Rab3A ^{+/+})			
mEPSC amplitude	14.2 ± 2.4 pA (6)	15.9 ± 1.9 pA (7)	+12.0	p = 0.25	15.4 ± 1.2 pA (7)	14.4 ± 0.8 pA (6)	-6.5	p = 0.20
VGLUT1 site size	0.86 ± 0.06 μm ² (10)	0.79 ± 0.04 μm ² (10)	-8.1	p = 0.11	0.85 ± 0.04 μm ² (10)	1.05 ± 0.07 μm ² (10)	+23.5	p = 0.15
VGLUT1 site intensity	919 ± 35 a.u. (10)	832 ± 18 a.u. (10)	-9.5	p = 0.10	526 ± 17 a.u. (10)	629 ± 33 a.u. (10)	+19.6	p = 0.10
VGLUT1 site integral	445647 ± 39114 a.u. (10)	355055 ± 22354 a.u. (10)	-20.3	p = 0.07	233589 ± 16145 a.u. (10)	329851 ± 32856 a.u. (10)	+41.2	p = 0.10
Quantity	Rab3A ^{+/+} Culture #2				Rab3A ^{-/-} Culture #2 (not same culture as Rab3A ^{+/+})			
mEPSC amplitude	13.8 ± 2.2 pA (7)	17.7 ± 1.8 pA (8)	+28.2	p = 0.16	15.2 ± 1.8 pA (8)	13.5 ± 1.8 pA (9)	-11.2	p = 0.50
VGLUT1 site size	0.95 ± 0.09 μm ² (9)	1.01 ± 0.05 μm ² (9)	+6.3	p = 0.35	0.81 ± 0.05 μm ² (10)	0.82 ± 0.03 μm ² (9)	+1.2	p = 0.68
VGLUT1 site intensity	454 ± 19 a.u. (9)	459 ± 34 a.u. (9)	+1.1	p = 0.57	505 ± 14 a.u. (10)	538 ± 19 a.u. (9)	+6.1	p = 0.29
VGLUT1 site integral	239671 ± 29790 a.u. (9)	253222 ± 35117 a.u. (9)	+5.6	p = 0.96	213066 ± 19080 a.u. (10)	236927 ± 12964 a.u. (9)	+11.2	p = 0.41
Quantity	Rab3A ^{+/+} Culture #3				Rab3A ^{-/-} Culture #3 (not same culture as Rab3A ^{+/+})			
mEPSC amplitude	13.4 ± 0.8 pA (10)	15.7 ± 1.0 pA (9)	+17.2	p = 0.10	13.9 ± 1.2 pA (6)	12.9 ± 1.5 pA (6)	-7.2	p = 0.26
VGLUT1 site size	1.64 ± 0.13 μm ² (10)	1.41 ± 0.10 μm ² (10)	-13.4	p = 0.17	1.01 ± 0.05 μm ² (10)	0.97 ± 0.07 μm ² (10)	-4.0	p = 0.76
VGLUT1 site intensity	700 ± 33 a.u. (10)	608 ± 12 a.u. (10)	-13.1	*p = 0.02	568 ± 31 a.u. (10)	538 ± 35 a.u. (10)	-5.3	p = 0.55
VGLUT1 site integral	587932 ± 59742 a.u. (10)	436840 ± 31327 a.u. (10)	-25.7	*p = 0.05	324823 ± 42586 a.u. (10)	297499 ± 37637 a.u. (10)	-8.4	p = 0.82

Regarding mEPSC frequency, in cultures with Rab3A^{+/+} neurons, frequency trended upward after activity blockade, whereas in the culture with Rab3A^{-/-} neurons, frequency was already increased, and trended downward after activity blockade.

In summary, in the absence of Rab3A from glia, the homeostatic plasticity of mEPSC amplitude following activity blockade was completely normal, whereas in the absence of Rab3A in neurons, homeostatic plasticity was greatly diminished. This result makes it highly unlikely that Rab3A is required for the release of TNF α , or another factor from glia, that induces a homeostatic upregulation of postsynaptic receptors and thereby increases mEPSC amplitude following TTX treatment. Neuronal Rab3A appears to mediate the homeostatic increase in mEPSC amplitude following activity blockade.

Discussion

We found that homeostatic synaptic plasticity of mEPSC amplitude in dissociated mixed cultures of mouse cortical neurons and glia behaved remarkably similar to the mouse NMJ in response to loss of Rab3A function: in cultures from Rab3A^{-/-} mice, the increase in mEPSC amplitudes following prolonged network silencing by TTX was strongly diminished, and in cultures from Rab3A^{Ebd/Ebd} mice, basal mEPSC amplitude was increased compared to that of wild-type cultures and was not further modified by TTX treatment. These results suggest that normal function of the synaptic vesicle protein Rab3A is required for the homeostatic increase of mEPSC amplitude in cortical cultures. The effect of TTX on mEPSC amplitude was not due to increases in Ca²⁺-permeable homomers because acute NASPM application was unable to reverse the increase in amplitude after TTX treatment. We used imaging of immunofluorescence to demonstrate a strong trend towards increased GluA2 levels at synapses in activity-blocked cultures from Rab3A^{+/+} mice that was no longer apparent in the absence of Rab3A, suggesting the surprising idea that this protein, previously implicated primarily in the regulation of calcium-dependent vesicle fusion, is also important for the regulation of surface expression of postsynaptic receptors. We found a significant increase in mEPSC frequency after activity blockade that was no longer significant in the absence of Rab3A, which was less surprising as frequency is known to depend on presynaptic factors. At the neuromuscular junction, higher mEPC frequencies that occur in 0.5 mM extracellular calcium and above were significantly reduced in the Rab3A deletion mouse (Coleman et al., 2007). On the other hand, the increase in frequency after TTX treatment is also strongly diminished after acute application of NASPM, suggesting there may be an increase in the number of synapses with only Ca²⁺-permeable homomers. It remains to be determined if this effect is Rab3A-dependent or if there are two distinct mechanisms contributing to homeostatic plasticity of frequency.

The levels of the glutamate transporter, VGLUT1, were not increased after activity blockade under our experimental conditions. A similar lack of increase in VGLUT1 levels, relative to synapsin in imaging of immunofluorescence, was previously observed in hippocampal cultures treated with NBQX, an AMPA receptor blocker (see supplemental data, (Wilson et al., 2005)). On the other hand, treatment with AP5, an NMDA receptor blocker, was able to induce an increase in VGLUT1 levels relative to synapsin (Wilson et al., 2005), suggesting that this presynaptic response depends on the type of activity manipulation. Another study showed increased VGLUT1 mRNA and protein in western blots in cortical cultures treated with TTX, but only examined VGLUT1 at synapses using imaging following treatment with bicuculline (De Gois et al., 2005). Taken together with our findings, it appears that there is not strong evidence that the amount of VGLUT1 at synapses is a contributing factor to the increase in mEPSC amplitude after activity blockade with TTX.

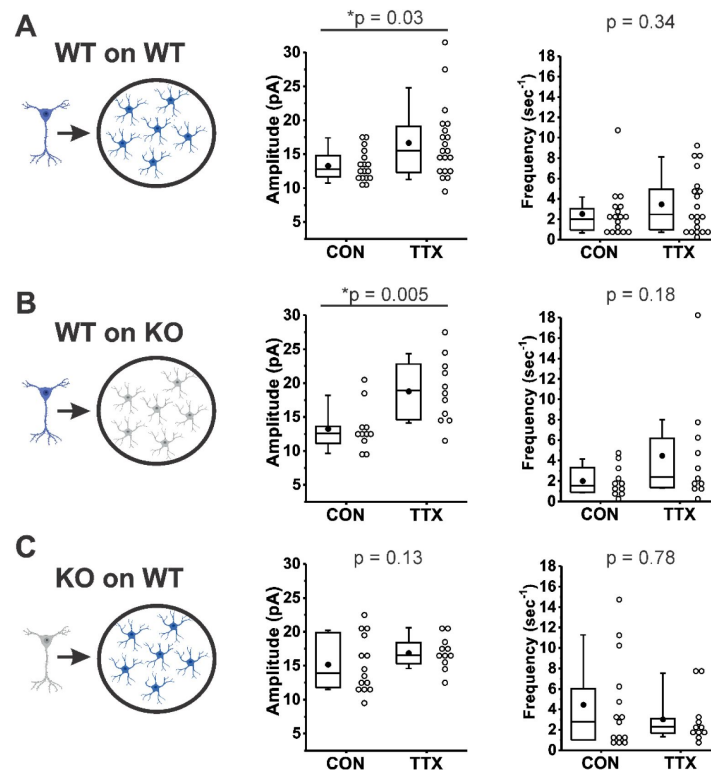


Figure 8.

Rab3A in neurons, not astrocytes, was required for full TTX-induced homeostatic plasticity.

(A)-(C), mEPSC amplitude (Middle) and frequency (Right) data from dissociated cortical neurons plated on an astrocyte feeder layer, each prepared separately from the type of mice depicted in the schema (Left): (A) Neurons from Rab3A^{+/+} mice plated on astrocytes from Rab3A^{+/+} mice.

Box plots for mEPSC amplitudes (CON, N = 17 cells, 13.3 ± 0.5 pA; TTX, N = 20 cells, 16.7 ± 1.2 pA) and mEPSC frequency in the same recordings (mean, CON, 2.54 ± 0.57 sec⁻¹; TTX, 3.48 ± 0.64 sec⁻¹); from 4 cultures. (B) Neurons from Rab3A^{+/+} mice plated on astrocytes from Rab3A^{-/-} mice. Box plots for average mEPSC amplitude (CON, N = 11 cells, 13.3 ± 1.0 pA; TTX, N = 11 cells, 18.8 ± 1.4 pA) and average mEPSC frequency in the same recordings (CON, 2.01 ± 0.41 sec⁻¹; TTX, 4.47 ± 1.53 sec⁻¹); from 2 cultures. (C) Neurons from Rab3A^{-/-} neurons plated on astrocytes from Rab3A^{+/+} mice. Box plots for average mEPSC amplitude (CON, N = 14 cells, 15.2 ± 1.1 pA; TTX, N = 11 cells, 16.9 ± 0.7 pA) and mEPSC frequency in the same recordings (CON, 4.47 ± 1.21 sec⁻¹; TTX, 3.02 ± 0.70 sec⁻¹); from 3 cultures. Box plot parameters: box ends, 25th and 75th percentiles; whiskers, 10th and 90th percentiles; open circles represent means from individual cells; line, median; dot, mean. p values (shown on the graphs) are from Kruskal-Wallis test. p-values denoted with * and underline indicates significance of $p < 0.05$.

Homeostatic plasticity of postsynaptic receptors was less consistent than that of mEPSC amplitude

While we found that the effect of TTX treatment on mEPSC amplitude and GluA2 receptor levels were of very similar magnitude, the effect on GluA2 receptor levels was not statistically significant for the pooled data sets acquired in the same cultures. We determined that this discrepancy was due to one of the three cultures showing a downward trend in GluA2 receptor levels after activity blockade. In light of the extensive literature documenting increases in GluA1 or GluA2 receptors, or both, after activity blockade, the less than convincing increase in GluA2 receptor levels in the current work came as a surprise. Therefore we reviewed what has previously been shown, and were able to identify only 5 studies that examined the effect of activity blockade using TTX alone, reported levels of AMPA receptors specifically localized to synapses, and recorded mEPSCs in the same study, although not in the same cultures (Hou et al., 2008 [DOI](#); Ibata et al., 2008 [DOI](#); Jakawich et al., 2010a [DOI](#); Xu and Pozzo-Miller, 2017 [DOI](#); Dubes et al., 2022 [DOI](#)). Since our issue is not with the magnitude of the changes being different for receptors and mEPSC amplitudes, but with statistical significance (or consistency) being different between the two methods, we wanted to determine whether the previous studies showed any evidence of greater variability in receptor measurements. Only a single study had comparable sample sizes for receptors and mEPSC amplitudes (either because number of receptor sites is used instead of cells, or because the number of cells differs for the two types of experiments), and the p value was < 0.01 for mEPSC amplitudes ($n = 9$ and 10 cells for CON and TTX, respectively), but < 0.05 for receptor data ($n = 13$ and 14 cells for CON and TTX) (Xu and Pozzo-Miller, 2017 [DOI](#)). This single example is consistent with the idea that the receptor data are more variable than mEPSC data. Even if a difference in consistency could be established in multiple studies, it remains a possibility that the disparity is attributable to the unique limitations of the two types of experiments. Nevertheless, the finding that mEPSC amplitude can increase despite a decrease in receptors suggests receptors may not be the only contributor to the TTX effect on mEPSC amplitude.

Neuronal, not glial, Rab3A may be required for the homeostatic release of a signaling molecule

Our results support the surprising possibility that the presynaptic vesicle protein Rab3A regulates postsynaptic AMPARs. Astrocytic release of TNF α was shown to mediate the increase in mEPSC amplitudes in prolonged TTX-treated cultures of dissociated hippocampal neurons and cultures of hippocampal slices (Stellwagen and Malenka, 2006 [DOI](#); Heir et al., 2024 [DOI](#)), but we found that loss of Rab3A in glia did not disrupt the increase in mEPSC amplitude after activity blockade. Further, we previously showed that the homeostatic increase in NMJ mEPSC amplitude was completely normal in the absence of TNF α (Wang et al., 2011 [DOI](#)). This leaves neuronal Rab3A as the potential controller of a soluble factor regulating surface expression of receptors.

One possible signaling molecule being controlled by Rab3A is brain-derived neurotrophic factor (BDNF). Addition of exogenous BDNF to neuronal cultures prevents the increase in mEPSC amplitude following activity blockade (Rutherford et al., 1998 [DOI](#); Benevento et al., 2016 [DOI](#)) but see (Smith-Dijak et al., 2019 [DOI](#)), and BDNF mRNA is reduced after activity blockade in vitro (Benevento et al., 2016 [DOI](#); Miyasaka and Yamamoto, 2021 [DOI](#)) and in vivo (Castren et al., 1992 [DOI](#)). Also, after reduction of secreted BDNF in culture media via the BDNF scavengers TrkB-FC or TrkB-IgG, mEPSC amplitudes are increased (Rutherford et al., 1998 [DOI](#); Benevento et al., 2016 [DOI](#); Smith-Dijak et al., 2019 [DOI](#)). Reduction of Rab3A in astrocytes decreases astrocytic BDNF release (Hong et al., 2016 [DOI](#)). Taken together, these studies suggest that activity blockade could cause a Rab3A-dependent inhibition of BDNF release, leading to an increase in postsynaptic receptors. However, we have no evidence that Rab3A is required for the regulation of BDNF release from neurons, so it is just as likely that after activity blockade, Rab3A acts in a positive way to promote release of a trophic molecule that stimulates the insertion of postsynaptic receptors. It is unlikely that Rab3A is

required when activity is normal because receptor levels do not change in the same way with loss of Rab3A as they do with loss of activity. Rather, our data fit with a model in which Rab3A mainly becomes engaged in response to activity blockade (**Figure 9**).

Is Rab3A acting in the postsynaptic cell to cause homeostatic increase in mEPSC amplitude?

To our knowledge, there is no evidence localizing Rab3A to postsynaptic sites. Other presynaptic molecules, such as SNARE proteins, synaptotagmins, and NSF, have been identified in the postsynaptic compartment (Lledo et al., 1998; Nishimune et al., 1998; Osten et al., 1998; Song et al., 1998; Araki et al., 2010; Kennedy et al., 2010; Suh et al., 2010; Jurado et al., 2013; Hussain and Davanger, 2015; Gu et al., 2016; Wu et al., 2017) but Rab3A localization was not examined in those studies. On the other hand, a recent report implicates Rab3A in the localization of plasma membrane proteins in rafts in HEK cells and Jurkat-T cells, indicating Rab3A can be found in locations other than the presynaptic nerve terminal (Diaz-Rohrer et al., 2023). In this study a reduction in all Rab3 isoforms led to an impairment of plasma membrane localization of epidermal growth factor receptor (EGFR), so it is possible that Rab3A is acting similarly in postsynaptic dendrites to promote transport of GluA2 receptors to the surface after activity blockade (**Figure 9**). Another possibility, not depicted in **Figure 9**, is that Rab3A acts postsynaptically to regulate the release of a trophic factor that in turn regulates surface expression of GluA2 receptors. This idea is consistent with evidence that BDNF is synthesized and released locally from dendrites (Jakawich et al., 2010b; Thapliyal et al., 2022).

Alternate contributors to homeostatic increase in mEPSC amplitude

We do not think Rab3A impacts homeostatic synaptic plasticity solely through pre- or postsynaptic regulation of GluA2 receptor trafficking to the plasma membrane. First, prolonged activity blockade with TTX at the NMJ led to an increase in mEPSC amplitude that was not accompanied by any increase in AChR levels, yet it was dependent on Rab3A (Wang et al., 2011). Second, mEPSC amplitudes increased in three cortical cultures treated with TTX, but GluA2 receptor levels in one of the cultures actually went down. An increase in conductance could explain the increase in mEPSC amplitudes in the absence of receptor level increases, but all evidence to date suggests that the only way to increase conductance is to either switch from GluA2-containing heteromeric receptors to GluA1 homomers, which have much greater conductance (Oh and Derkach, 2005; Soares et al., 2013; Hou et al., 2015; Xu and Pozzo-Miller, 2017; Benke and Traynelis, 2019; Silva et al., 2019; Dubes et al., 2022), or, to post-translationally modify the receptor. Such conductance-altering modifications have been demonstrated only for GluA1 (Derkach et al., 1999; Kristensen et al., 2011; Jenkins and Traynelis, 2012). Importantly, any change in conductance due to an increase in GluA1 homomers or modification of GluA1 should have been detected by NASPM being able to disrupt the homeostatic increase in mEPSC amplitude, and this was not what we observed.

The other contributor to mEPSC amplitude besides receptors is the size of the presynaptic quantum, or the amount of transmitter released during a single fusion event. As stated above, we found a downward trend in VGLUT1 levels after activity blockade. This downward trend was not apparent in cultures prepared from Rab3A^{-/-} mice, suggesting a possible Rab3A-dependent effect. Interestingly, Rab3A levels are reduced 50% in hippocampal extracts from mice lacking VGLUT1 (Freneau et al., 2004). Coupled with our finding that loss of Rab3A was associated with a highly significant decrease in VGLUT1 levels in untreated cultures, these results suggest an interdependence between Rab3A and VGLUT1. More importantly, our results rule out VGLUT1 levels as the source for increased mEPSC amplitude.

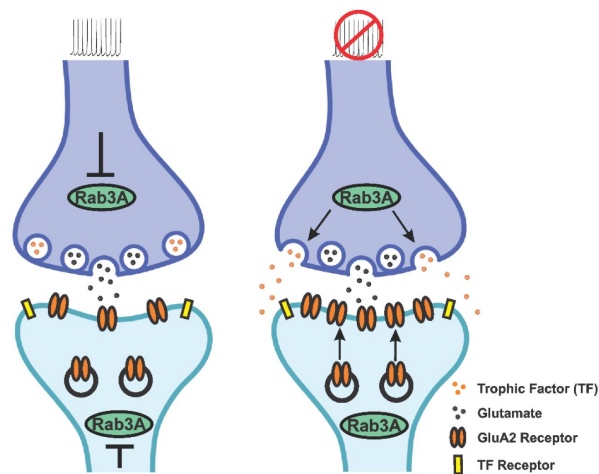


Figure 9.

Model depicting the requirement for neuronal Rab3A in the homeostatic increase in synaptic GluA2 receptor levels after treatment with TTX.

Under normal activity levels (Left), neither presynaptic or postsynaptic Rab3A is engaged in the regulation of GluA2 receptor levels. After prolonged activity blockade (Right), Rab3A located in the presynaptic terminal promotes the release of a trophic factor (orange dots) that diffuses across the synapse and facilitates transport of postsynaptic vesicles containing GluA2 AMPA receptors (dark orange channels) to the surface membrane. Rab3A located in the postsynaptic dendrite directly promotes fusion of vesicles containing GluA2 AMPA receptors. Fusion of presynaptic vesicles containing glutamate (black dots) causes mEPSCs when glutamate binds to AMPA receptors. Adapted from "Synaptic Cleft" template, by BioRender.com (2024).

The amount of transmitter released during vesicle fusion can be affected by the kinetics of the fusion pore opening (Chang et al., 2017 [↗](#)). We previously used amperometric measurements in mouse adrenal chromaffin cells to show loss of Rab3A increased the occurrence of very small amplitude fusion pore feet (Wang et al., 2008 [↗](#)). Paired with our finding that there is an increase in the occurrence of very slow-rising, abnormally shaped mEPSCs at the neuromuscular junction of Rab3A^{-/-} mice, our data suggest that small synaptic vesicles may have a fusion pore step under certain circumstances (see also (Chiang et al., 2018 [↗](#))). If activity blockade causes an increase in mEPSC amplitude due to a more rapid opening of a fusion pore, or a larger fusion pore conductance, it is possible that Rab3A is required for this modulation.

Finally, a simple way to increase the amount of transmitter released by a vesicle is to enlarge vesicle size. Larger vesicle diameter has been observed following activity disruption in multiple paradigms, including ocular deprivation, cortical ablation, and immobilization (Kawana et al., 1971 [↗](#); Lloret and Saavedra, 1975 [↗](#); Cheresnarov et al., 1978 [↗](#)). Loss of Rab3 family members is associated with increased vesicle size (Riedel et al., 2002 [↗](#); Schonn et al., 2010 [↗](#)). Finally, an increase in volume has been shown to increase miniature endplate junctional current amplitude at the *Drosophila* NMJ (Karunanithi et al., 2002 [↗](#)). Rab3A could be required for an activity-dependent increase in vesicle size, but the increase in diameter needed for a 25% increase in transmitter, and therefore mEPSC amplitude, would be difficult to detect.

To our knowledge, this is the first evidence of a presynaptic protein being implicated in the homeostatic regulation of mEPSC amplitude and AMPARs in neuronal cultures. Homeostatic synaptic plasticity is becoming increasingly implicated in essential normal functions such as sleep (Tononi and Cirelli, 2014 [↗](#); Diering et al., 2017 [↗](#); Torrado Pacheco et al., 2021 [↗](#)), and pathological conditions such as epilepsy, neuropsychiatric disorders, Huntington's, and alcohol use disorder (Trasande and Ramirez, 2007 [↗](#); Fernandes and Carvalho, 2016 [↗](#); Wang et al., 2017 [↗](#); Lovinger and Abrahao, 2018 [↗](#); Smith-Dijak et al., 2019 [↗](#); Lignani et al., 2020 [↗](#); Suzuki et al., 2021 [↗](#); Kavalali and Monteggia, 2023 [↗](#)). Therefore our identification of a novel regulatory pathway involving Rab3A may have important therapeutic implications. In future work it will be important to determine whether Rab3A is acting presynaptically or postsynaptically to regulate GluA2 receptors, and what other contributor to mEPSC amplitude depends on Rab3A, possibly fusion pore characteristics or vesicle volume.

Materials and methods

Animals. To determine the role of Rab3A in homeostatic plasticity in mouse cortical cultures, we employed two distinct genetic mouse strains with altered Rab3A function. Rab3A deletion mice were generated as follows: Rab3A^{+/-} heterozygous mice were bred and genotyped as previously described (Kapfhamer et al., 2002 [↗](#); Wang et al., 2008 [↗](#)), and maintained as a colony using multiple heterozygous breeding pairs. Homozygous Rab3A^{+/+} or Rab3A^{-/-} mouse pups were obtained by breeding Rab3A^{+/+} pairs and Rab3A^{-/-} pairs, respectively, so that the pups were the first generation progeny of a homozygous mating. This protocol minimizes the tendency of multiple generations of Rab3A^{+/+} and Rab3A^{-/-} homozygous breedings to produce progeny that are more and more genetically distinct. Rab3A^{Ebd/Ebd} mice were identified in an EU-mutagenesis screen of C57BL/6J mice, and after a cross to C3H/HeJ, were backcrossed for 3 generations to C57BL/6J (Kapfhamer et al., 2002 [↗](#)). Rab3A^{+Ebd} heterozygous mice were bred at Wright State University and genotyped in a two-step procedure: 1. a PCR reaction (forward primer: TGA CTC CTT CAC TCC AGC CT; reverse primer: TGC ACT GCA TTA AAT GAC TCC T) followed by 2. a restriction digest with enzyme Bsp1286I (New England Biolabs) that distinguishes the Earlybird mutant by its different base-pair products. Rab3A^{+/-} mice were backcrossed with Rab3A^{+/+} mice from the Earlybird heterozygous colony for 11 generations in an attempt to establish a single wild

type strain, but differences in mEPSC amplitude and adrenal chromaffin cell calcium currents persisted, likely due to genes that are close to the Rab3A site, resulting in two wild type strains: 1. Rab3A^{+/+} from the Rab3A^{+/-} colony, and 2. Rab3A^{+/+} from the Rab3A^{+Ebd} colony.

Primary Culture of Mouse Cortical Neurons. Primary dissociated cultures of mixed neuronal and glia populations were prepared as previously described (Hanes et al., 2020 [DOI](#)). Briefly, postnatal day 0-2 (P0-P2) Rab3A^{+/+}, Rab3A^{-/-} or Rab3A^{Ebd/Ebd} neonates were euthanized by rapid decapitation, as approved by the Wright State University Institutional Animal Care and Use Committee, and brains were quickly removed. Each culture was prepared from the cortices harvested from two animals; neonates were not sexed. Cortices were collected in chilled Neurobasal-A media (Gibco) with osmolarity adjusted to 270 mOsm and supplemented with 40 U/ml DNase I (ThermoFisher Scientific). The tissues were digested with papain (Worthington Biochemical) at 20 U/ml at 37°C for 20 minutes followed by trituration with a sterile, fire-polished Pasteur pipette, then filtered through a 100 µm cell strainer, and centrifuged at 1100 rpm for 2 minutes. After discarding the supernatant, the pellet was resuspended in room temperature Neurobasal-A media (270 mOsm), supplemented with 5% fetal bovine serum for glia growth, and 2% B-27 supplement to promote neuronal growth (Gibco), L-glutamine, and gentamicin (ThermoFisher Scientific). Neurons were counted and plated at 0.15×10^6 cells/coverslip onto 12 mm coverslips pre-coated with poly-L-lysine (BioCoat, Corning). This plating density produces a “high density” culture characterized by a complex mesh of neuronal processes criss-crossing the field of view, completely filling the space between cell bodies (see **Figure 4** [DOI](#)). We found in preliminary experiments that an increase in mEPSC amplitude in mouse cortical cultures was inconsistent when cell density was low, likely due to lower levels of baseline activity, although it is a limitation of this study that we did not directly measure activity levels in cultures prepared from Rab3A^{+/+}, Rab3A^{-/-} or Rab3A^{Ebd/Ebd} mice, and therefore do not know if differences in results are due to differences in activity levels. While unlikely for Rab3A^{+/+} and Rab3A^{-/-} studies, given the mEPSC amplitude in untreated cultures that was relatively small, it remains possible that in Rab3A^{Ebd/Ebd} studies, the larger mEPSC amplitude in untreated cultures was due to a loss of activity in these cultures. The culture media for the first day (0 DIV) was the same as the above Neurobasal-A media supplemented with FBS, B-27, L-glutamine, and gentamicin, and was switched after 24 hours (1 DIV) to media consisting of Neurobasal-A (270 mOsm), 2% B-27, and L-glutamine without FBS to avoid its toxic effects on neuronal viability and health (Stellwagen and Malenka, 2006 [DOI](#)). Half of the media was changed twice weekly and experiments were performed at 13-14 DIV. Two days prior to experiments, tetrodotoxin (TTX) (500 nM; Tocris), a potent Na⁺ channel blocker, was added to some cultures to chronically silence all network activity and induce homeostatic synaptic plasticity mechanisms, while untreated sister cultures served as controls. Cultures prepared from mutant mice were compared with cultures from wild-type mice from their respective colonies. Note that the cultures comprising the Rab3A^{+/+} data here are a subset of the data previously published in Hanes et al., 2020 [DOI](#). This smaller data set was restricted to the time period over which cultures were also prepared from Rab3A^{-/-} mice.

Preparation of Glial Feeder Layers. Glial feeder layers were prepared from the cortices of P0-P2 Rab3A^{+/+} or Rab3A^{-/-} mouse pups as described previously (Stellwagen and Malenka, 2006 [DOI](#)). Briefly, cortices were dissected and cells were dissociated as described above. Cell suspensions of mixed neuronal and glial populations were plated onto glass coverslips pre-coated with poly-L-lysine in Dulbecco's Modified Eagle Media (ThermoFisher Scientific) supplemented with 5% FBS (to promote glial proliferation and to kill neurons), L-glutamine, and gentamicin, and maintained in an incubator at 37°C, 5% CO₂; cultures were maintained in this manner for up to 1 month to generate purely glial cultures (all neurons typically died off by 7 DIV). Culture media was replaced after 24 hours, and subsequent media changes were made twice weekly, replacing half of the culture media with fresh media. Feeder layers were not used for neuronal seeding until all native neurons were gone and glia approached 100% confluency (visually inspected).

Plating of Neurons on Glial Feeder Layers. Cortical neurons were obtained as described above. The cell pellet obtained was resuspended in Neurobasal-A (osmolarity adjusted to 270 mOsm) containing B27 (2%, to promote neuronal growth), L-glutamine, and 5-fluorodeoxyuridine (FdU, a mitotic inhibitor; Sigma). Addition of FdU was used to prevent glial proliferation and contamination of the feeder layer with new glia, promoting only neuronal growth on the feeder layers (FdU-containing media was used for the maintenance of these cultures and all subsequent media changes). Glial culture media was removed from the feeder layer cultures, and the neuronal cell suspension was plated onto the glial feeder cultures. The culture strategy used to distinguish the relative roles of neuronal and glial Rab3A is outlined in **Figure 8**, Left. At 1 DIV, all of the culture media was removed and replaced with fresh Neurobasal-A media containing FdU described above, and half of the media was replaced twice per week for all subsequent media changes. Cultures were maintained in a 37°C, 5% CO₂ incubator for 13-14 DIV.

Whole-Cell Voltage Clamp to Record mEPSCs. At 13-14 DIV, mEPSCs from TTX-treated and untreated sister cultures of Rab3A^{+/+} or Rab3A^{-/-} neurons from the Rab3A^{+/+} colony, or Rab3A^{+/+} or Rab3A^{Ebd/Ebd} neurons from the Rab3A^{+/Ebd} colony, were recorded via whole-cell voltage clamp to assess the role of Rab3A in homeostatic synaptic plasticity. Recordings were taken from pyramidal neurons, which were identified visually by a prominent apical dendrite; images were taken of all cells recorded from. Cells were continuously perfused with a solution consisting of (in mM): NaCl (115), KCl (5), CaCl₂ (2.5), MgCl₂ (1.3), dextrose (23), sucrose (26), HEPES (4.2), pH = 7.2 (Stellwagen and Malenka, 2006). On the day of recording, the osmolarity of the media from the cultures was measured (normally 285 – 295 mOsm) and the perfusate osmolarity was adjusted to match the culture osmolarity, to protect against osmotic shock to the neurons. To isolate glutamatergic mEPSCs, TTX (500 nM) and picrotoxin (50 μM) were included in the perfusion solution to block action potentials and GABAergic currents, respectively. The NMDA receptor antagonist, APV, was not included in the perfusion solution because in pilot studies, all mEPSCs were blocked by 10 μM CNQX and 50 μM picrotoxin, demonstrating no APV-sensitive mEPSCs were present (data not shown). Previous studies have described NMDA mEPSCs in neuronal cultures, but recordings were performed in 0 Mg²⁺ (Watt et al., 2000; Sutton et al., 2006). In our recordings, the presence of extracellular Mg²⁺ (1.3 mM) and TTX would mean that the majority of NMDA receptors are blocked by extracellular Mg²⁺ at resting V_m. Patch electrodes (3.5 – 5 MΩ) were filled with an internal solution containing (in mM): K-gluconate (130), NaCl (10), EGTA (1), CaCl₂ (0.132), MgCl₂ (2), HEPES (10), pH = 7.2. Osmolarity was adjusted to 10 mOsm less than the perfusion solution osmolarity, which we have found improves success rate of achieving low access resistance recordings. Neurons were clamped at a voltage of -60 mV using an Axopatch 200B patch-clamp (Axon Instruments), recorded from for 2 – 5 minutes, and data were collected with Clampex 10.0/10.2 (Axon Instruments).

NASPM application. The antagonist of Ca²⁺-permeable AMPA receptors, N-naphthyl acetylspermine (NASPM, 20 μM; Tocris), was applied during recordings in a subset of experiments. NASPM is a synthetic analog of Joro Spider Toxin (JSTX) (Koike et al., 1997), and chemicals in this family block Ca²⁺ permeable glutamate receptors (Blaschke et al., 1993; Herlitze et al., 1993; Iino et al., 1996). The presence of the edited GluA2 subunit disrupts both the Ca²⁺ permeability and the sensitivity to JSTX and related substances (Blaschke et al., 1993; Bochet et al., 1994; Jonas and Burnashev, 1995). Therefore, 20 μM NASPM is expected to completely block AMPA receptors containing GluA1, GluA3 or GluA4 subunits (Hollmann et al., 1991; Burnashev et al., 1992), or the Kainate receptors containing GluA5 or GluA6 (Koike et al., 1997; Sun et al., 2009), but be ineffective against any heteromer or homomer containing GluA2 subunits. Because the effect of the spider toxins and their analogues are use-dependent (Herlitze et al., 1993; Iino et al., 1996; Koike et al., 1997), NASPM was applied with a mildly depolarizing high K⁺ solution (25 mM KCl, 95 mM NaCl) which we expected to trigger release of glutamate and opening of glutamate-activated receptors, allowing entry of the pore blocker. Baseline recordings were performed for 2 minutes in our standard perfusate, then were suspended while NASPM + High K⁺ solution was applied for 45 seconds, followed by a NASPM only solution for 5 minutes, after which recording

was recommenced for 5 minutes (because we found in pilot experiments that frequency was reduced following NASPM application). *Analysis of mEPSCs.* Miniature excitatory postsynaptic currents (mEPSCs) were manually selected using Mini Analysis software (Synaptosoft), a standard program used for mEPSC event detection and analysis. Records were filtered at 2 kHz using a low-pass Butterworth filter prior to selection. The program threshold was set at 3 pA but the smallest mEPSC selected was 4.01 pA. MiniAnalysis selects many false positives with the automated feature when a small threshold amplitude value is employed, due to random fluctuations in noise, so manual re-evaluation of the automated process is necessary to eliminate false positives. If the threshold value is set high, there are few false positives but small amplitude events that visually are clearly mEPSCs are missed, and manual re-evaluation is necessary to add back false negatives or the population ends up biased towards large mEPSC amplitudes. As soon as there is a manual step, bias is introduced. Interestingly, a manual reevaluation step was applied in a recent study that describes their process as ‘unbiased’ (Wu et al., 2020 [DOI](#)). In sum, we do not believe it is currently possible to perform a completely unbiased detection process. A fully manual detection process means that the same criterion (“does this *look* like an mEPSC?”) is applied to all events, not just the false positives, or the false negatives, which prevents the bias from being primarily at one end or the other of the range of mEPSC amplitudes. It is important to note that when performing the MiniAnalysis process, the researcher did not know whether a record was from an untreated cell or a TTX-treated cell.

Immunofluorescence, microscopy, and data analysis. Primary cultures of mouse cortical neurons were grown for 13-14 DIV. Antibodies to GluA2 (mouse ab against N-terminal, EMD Millipore) were added directly to live cultures at 1:40 dilution, and incubated at 37 °C in a CO₂ incubator for 45 minutes. Cultures were rinsed 3 times with PBS/5% donkey serum before being fixed with 4% paraformaldehyde. After 3 rinses in PBS/5% donkey serum, cultures were incubated in CY3-labeled donkey-anti-mouse secondary antibodies for 1 hour at room temperature, rinsed in PBS/5% donkey serum, permeabilized with 0.2% saponin, and incubated in chick anti-MAP2 (1:2500, AbCAM) and rabbit anti-VGLUT1 (1:4000, Synaptic Systems) for 1 hour at room temperature in PBS/5% donkey serum. After rinsing with PBS/5% donkey serum, coverslips were incubated with 488-anti chick and CY5-anti rabbit secondary antibodies for 1 hour at room temperature, rinsed, blotted to remove excess liquid, and inverted on a drop of Vectashield (Vector Labs). Coverslips were sealed with nail polish and stored at 4 °C for < 1 week before imaging. All secondary antibodies were from Jackson ImmunoResearch and were used at 1:225 dilution.

Coverslips were viewed on a Fluoview FV1000 laser scanning confocal microscope with a 60x oil immersion, 1.35 NA objective. Once a pyramidal neuron was identified and a single confocal section of the cell acquired (see Top, **Figure 4** [DOI](#)), Fluoview 2.1 software was used to zoom in on the primary dendrite (5X) and a series of confocal sections were taken every 0.5 μm. Images were analyzed offline with ImagePro 6 (Cybernetics). The composite image was used to locate synaptic sites containing both VGLUT1 and GluA2 immunoreactivity in close apposition to each other and to the primary dendrite or a secondary branch identified with MAP2 immunoreactivity. An area of interest (AOI) was manually drawn around the GluA2 cluster in the confocal section in which it was the brightest. The AOIs for a dendrite were saved in a single file; the AOI number and the confocal section it was associated with were noted for later retrieval. For quantification, AOIs were loaded, an individual AOI was called up on the appropriate section, and the count/measurement tool used to apply a threshold. Multiple thresholds were used, 400-550, but we found that within a given experiment, the magnitude of the TTX effect was not altered by different thresholds, therefore, all data presented here use a threshold of 450. Pixels within the cluster that were above the threshold were automatically outlined, and size, average intensity, and integral of the outlined region saved to Origin2020 (OriginLab) file for calculating mean values. For analyzing VGLUT1 sites, previous AOIs were examined, and if not directly over the VGLUT1 site, moved to encompass it (this happened when GluA2 and VGLUT1 were side by side rather than overlapping). The threshold was 200 for all VGLUT1 data except Rab3A^{+/+} Culture #1, where threshold was 400.

There was no threshold that worked for both that culture and the other cultures, with too much of the background regions being included for Culture #1 with a threshold of 200, and too little of the synaptic site area being included for the other Cultures with a threshold of 400.

Statistics. In our previous publication (Hanes et al., 2020 [DOI](#)), we described the characteristics of homeostatic plasticity of mEPSC amplitudes in mouse cortical cultures using multiple mEPSC amplitude quantiles from each cell to create data sets with thousands of samples. We compared the cumulative distribution functions for mEPSC amplitudes in untreated vs. TTX-treated cultures, as well as sorting the values from smallest to largest, and plotted TTX values vs. control values in the rank order plot first described in (Turrigiano et al., 1998 [DOI](#)) to demonstrate synaptic scaling. Finally, we plotted the ratio of TTX values/control values vs. control values to show that the ratio, which represents the effect of TTX, increased with increasing control values, which we termed “divergent scaling.” Here, we wanted to avoid the inflation of sample size caused by pooling multiple mEPSCs per cell in order to statistically compare the effects of TTX in cultures of wild type mice, Rab3A^{-/-} mice and Rab3A^{Ebd/Ebd} mutant mice. Therefore, we computed the mean mEPSC amplitude for each cell and compared means from untreated and TTX-treated cultures with the non-parametric Kruskal-Wallis test, with N = the number of cells. The overall means are presented ± SEM in the text, and p values are displayed above each plot. The same process was carried out for the characteristics of synaptic GluA2 receptor clusters and VGLUT1 sites in imaging experiments, with N = the number of dendrites (one dendrite was sampled for each cell). Although we quantified three characteristics of the GluA2 clusters and VGLUT1 sites, size, average intensity, and integral, we have not corrected p values for multiple comparisons. Means are presented as box plots + data, with the box at 25 and 75 percentiles, whiskers at 10 and 90 percentiles, a dot indicating the mean, and a line indicating the median. In **Figure 3** [DOI](#), mEPSC amplitudes were measured within the same cell before and after NASPM application, so a paired T test was used to compare the pre- and post-NASPM mEPSC amplitudes. Statistics were computed and plots were created using Origin2020 (OriginLab).

Acknowledgements

The authors thank E. Whitlock and H. Ghouse for their contributions to confocal image analyses. This work was partially supported by NINDS P01 NS057228, Tim Cope, PI, KE, PI project 4.

References

1. Araki Y, Lin DT, Huganir RL (2010) **Plasma membrane insertion of the AMPA receptor GluA2 subunit is regulated by NSF binding and Q/R editing of the ion pore** *Proc Natl Acad Sci U S A* **107**:11080–11085
2. Benevento M *et al.* (2016) **Histone Methylation by the Kleefstra Syndrome Protein EHMT1 Mediates Homeostatic Synaptic Scaling** *Neuron* **91**:341–355
3. Benke T, Traynelis SF (2019) **AMPA-Type Glutamate Receptor Conductance Changes and Plasticity: Still a Lot of Noise** *Neurochem Res* **44**:539–548
4. Blaschke M, Keller BU, Rivosecchi R, Hollmann M, Heinemann S, Konnerth A (1993) **A single amino acid determines the subunit-specific spider toxin block of alpha-amino-3-hydroxy-5-methylisoxazole-4-propionate/kainate receptor channels** *Proc Natl Acad Sci U S A* **90**:6528–6532
5. Bochet P, Audinat E, Lambolez B, Crepel F, Rossier J, Iino M, Tsuzuki K, Ozawa S (1994) **Subunit composition at the single-cell level explains functional properties of a glutamate-gated channel** *Neuron* **12**:383–388
6. Burnashev N, Monyer H, Seeburg PH, Sakmann B (1992) **Divalent ion permeability of AMPA receptor channels is dominated by the edited form of a single subunit** *Neuron* **8**:189–198
7. Castillo PE, Janz R, Sudhof TC, Tzounopoulos T, Malenka RC, Nicoll RA (1997) **Rab3A is essential for mossy fibre long-term potentiation in the hippocampus** *Nature* **388**:590–593
8. Castren E, Zafra F, Thoenen H, Lindholm D (1992) **Light regulates expression of brain-derived neurotrophic factor mRNA in rat visual cortex** *Proc Natl Acad Sci U S A* **89**:9444–9448
9. Chang CW, Chiang CW, Jackson MB (2017) **Fusion pores and their control of neurotransmitter and hormone release** *J Gen Physiol* **149**:301–322
10. Cheresharov L, Ovtscharoff W, Manolov S (1978) **Ultrastructural changes in the anterior horn synapses of rat spinal cord under different locomotor conditions** *J Neural Transm* **42**:9–21
11. Chiang CW, Chang CW, Jackson MB (2018) **The Transmembrane Domain of Synaptobrevin Influences Neurotransmitter Flux through Synaptic Fusion Pores** *J Neurosci* **38**:7179–7191
12. Coleman WL, Bykhovskaia M (2009) **Rab3a-mediated vesicle recruitment regulates short-term plasticity at the mouse diaphragm synapse** *Mol Cell Neurosci* **41**:286–296
13. Coleman WL, Bykhovskaia M (2010) **Cooperative regulation of neurotransmitter release by Rab3a and synapsin II** *Mol Cell Neurosci* **44**:190–200
14. Coleman WL, Bill CA, Bykhovskaia M (2007) **Rab3a deletion reduces vesicle docking and transmitter release at the mouse diaphragm synapse** *Neuroscience* **148**:1–6

15. De Gois S, Schafer MK, Defamie N, Chen C, Ricci A, Weihe E, Varoqui H, Erickson JD (2005) **Homeostatic scaling of vesicular glutamate and GABA transporter expression in rat neocortical circuits** *J Neurosci* **25**:7121–7133
16. Derkach V, Barria A, Soderling TR (1999) **Ca²⁺/calmodulin-kinase II enhances channel conductance of alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate type glutamate receptors** *Proc Natl Acad Sci U S A* **96**:3269–3274
17. Diaz-Rohrer B, Castello-Serrano I, Chan SH, Wang HY, Shurer CR, Levental KR, Levental I (2023) **Rab3 mediates a pathway for endocytic sorting and plasma membrane recycling of ordered microdomains** *Proc Natl Acad Sci U S A* **120**
18. Diering GH, Nirujogi RS, Roth RH, Worley PF, Pandey A, Huganir RL (2017) **Homer1a drives homeostatic scaling-down of excitatory synapses during sleep** *Science* **355**:511–515
19. Dubes S, Soula A, Benquet S, Tessier B, Poujol C, Favereaux A, Thoumine O, Letellier M (2022) **miR-124-dependent tagging of synapses by synaptopodin enables input-specific homeostatic plasticity** *EMBO J* **41**
20. Fan D, Grooms SY, Araneda RC, Johnson AB, Dobrenis K, Kessler JA, Zukin RS (1999) **AMPA receptor protein expression and function in astrocytes cultured from hippocampus** *J Neurosci Res* **57**:557–571
21. Fernandes D, Carvalho AL (2016) **Mechanisms of homeostatic plasticity in the excitatory synapse** *J Neurochem* **139**:973–996
22. Fremeau RT Jr, Kam K, Qureshi T, Johnson J, Copenhagen DR, Storm-Mathisen J, Chaudhry FA, Nicoll RA, Edwards RH (2004) **Vesicular glutamate transporters 1 and 2 target to functionally distinct synaptic release sites** *Science* **304**:1815–1819
23. Geppert M, Bolshakov VY, Siegelbaum SA, Takei K, De Camilli P, Hammer RE, Sudhof TC (1994) **The role of Rab3A in neurotransmitter release** *Nature* **369**:493–497
24. Gu Y, Chiu SL, Liu B, Wu PH, Delannoy M, Lin DT, Wirtz D, Huganir RL (2016) **Differential vesicular sorting of AMPA and GABAA receptors** *Proc Natl Acad Sci U S A* **113**:E922–931
25. Hanes AL, Koesters AG, Fong MF, Altimimi HF, Stellwagen D, Wenner P, Engisch KL (2020) **Divergent Synaptic Scaling of Miniature EPSCs following Activity Blockade in Dissociated Neuronal Cultures** *J Neurosci* **40**:4090–4102
26. Heir R, Abbasi Z, Komal P, Altimimi HF, Franquin M, Moschou D, Chambon J, Stellwagen D (2024) **Astrocytes Are the Source of TNF Mediating Homeostatic Synaptic Plasticity** *J Neurosci* **44**
27. Herlitze S, Raditsch M, Ruppersberg JP, Jahn W, Monyer H, Schoepfer R, Witzemann V (1993) **Argitoxin detects molecular differences in AMPA receptor channels** *Neuron* **10**:1131–1140
28. Hollmann M, Hartley M, Heinemann S (1991) **Ca²⁺ permeability of KA-AMPA-gated glutamate receptor channels depends on subunit composition** *Science* **252**:851–853
29. Hong Y, Zhao T, Li XJ, Li S (2016) **Mutant Huntingtin Impairs BDNF Release from Astrocytes by Disrupting Conversion of Rab3a-GTP into Rab3a-GDP** *J Neurosci* **36**:8790–8801

30. Hou Q, Zhang D, Jarzylo L, Hugarir RL, Man HY (2008) **Homeostatic regulation of AMPA receptor expression at single hippocampal synapses** *Proc Natl Acad Sci U S A* **105**:775–780
31. Hou Q, Ruan H, Gilbert J, Wang G, Ma Q, Yao WD, Man HY (2015) **MicroRNA miR124 is required for the expression of homeostatic synaptic plasticity** *Nat Commun* **6**
32. Hu JH *et al.* (2010) **Homeostatic scaling requires group I mGluR activation mediated by Homer1a** *Neuron* **68**:1128–1142
33. Hussain S, Davanger S (2015) **Postsynaptic VAMP/Synaptobrevin Facilitates Differential Vesicle Trafficking of GluA1 and GluA2 AMPA Receptor Subunits** *PLoS One* **10**
34. Ibata K, Sun Q, Turrigiano GG (2008) **Rapid synaptic scaling induced by changes in postsynaptic firing** *Neuron* **57**:819–826
35. Iino M, Koike M, Isa T, Ozawa S (1996) **Voltage-dependent blockage of Ca(2+)-permeable AMPA receptors by joro spider toxin in cultured rat hippocampal neurones** *J Physiol* **496**:431–437
36. Jakawich SK, Neely RM, Djakovic SN, Patrick GN, Sutton MA (2010) **An essential postsynaptic role for the ubiquitin proteasome system in slow homeostatic synaptic plasticity in cultured hippocampal neurons** *Neuroscience* **171**:1016–1031
37. Jakawich SK, Nasser HB, Strong MJ, McCartney AJ, Perez AS, Rakesh N, Carruthers CJ, Sutton MA (2010) **Local presynaptic activity gates homeostatic changes in presynaptic function driven by dendritic BDNF synthesis** *Neuron* **68**:1143–1158
38. Jenkins MA, Traynelis SF (2012) **PKC phosphorylates GluA1-Ser831 to enhance AMPA receptor conductance** *Channels (Austin)* **6**:60–64
39. Jonas P, Burnashev N (1995) **Molecular mechanisms controlling calcium entry through AMPA-type glutamate receptor channels** *Neuron* **15**:987–990
40. Jurado S, Goswami D, Zhang Y, Molina AJ, Sudhof TC, Malenka RC (2013) **LTP requires a unique postsynaptic SNARE fusion machinery** *Neuron* **77**:542–558
41. Kapfhamer D, Valladares O, Sun Y, Nolan PM, Rux JJ, Arnold SE, Veasey SC, Bucan M (2002) **Mutations in Rab3a alter circadian period and homeostatic response to sleep loss in the mouse** *Nat Genet* **32**:290–295
42. Karunanithi S, Marin L, Wong K, Atwood HL (2002) **Quantal size and variation determined by vesicle size in normal and mutant Drosophila glutamatergic synapses** *J Neurosci* **22**:10267–10276
43. Kavalali ET, Monteggia LM (2023) **Rapid homeostatic plasticity and neuropsychiatric therapeutics** *Neuropsychopharmacology* **48**:54–60
44. Kawana E, Akert K, Bruppacher H (1971) **Enlargement of synaptic vesicles as an early sign of terminal degeneration in the rat caudate nucleus** *J Comp Neurol* **142**:297–307
45. Kennedy MJ, Davison IG, Robinson CG, Ehlers MD (2010) **Syntaxin-4 defines a domain for activity-dependent exocytosis in dendritic spines** *Cell* **141**:524–535

46. Koike M, Iino M, Ozawa S (1997) **Blocking effect of 1-naphthyl acetyl spermine on Ca(2+)-permeable AMPA receptors in cultured rat hippocampal neurons** *Neurosci Res* **29**:27–36
47. Kristensen AS, Jenkins MA, Banke TG, Schousboe A, Makino Y, Johnson RC, Huganir R, Traynelis SF (2011) **Mechanism of Ca²⁺/calmodulin-dependent kinase II regulation of AMPA receptor gating** *Nat Neurosci* **14**:727–735
48. Leenders AG, da Silva FH Lopes, Ghijsen WE, Verhage M (2001) **Rab3a is involved in transport of synaptic vesicles to the active zone in mouse brain nerve terminals** *Mol Biol Cell* **12**:3095–3102
49. Lignani G, Baldelli P, Marra V (2020) **Homeostatic Plasticity in Epilepsy** *Front Cell Neurosci* **14**
50. Lledo PM, Zhang X, Sudhof TC, Malenka RC, Nicoll RA (1998) **Postsynaptic membrane fusion and long-term potentiation** *Science* **279**:399–403
51. Lloret IL, Saavedra JP (1975) **Enlargement of synaptic vesicles in degenerating nerve endings: a comparison between cat and monkey** *J Neurocytol* **4**:1–6
52. Lonart G, Janz R, Johnson KM, Sudhof TC (1998) **Mechanism of action of rab3A in mossy fiber LTP** *Neuron* **21**:1141–1150
53. Lovinger DM, Abrahao KP (2018) **Synaptic plasticity mechanisms common to learning and alcohol use disorder** *Learn Mem* **25**:425–434
54. Maienschein V, Marxen M, Volkhardt W, Zimmermann H (1999) **A plethora of presynaptic proteins associated with ATP-storing organelles in cultured astrocytes** *Glia* **26**:233–244
55. Miyasaka Y, Yamamoto N (2021) **Neuronal Activity Patterns Regulate Brain-Derived Neurotrophic Factor Expression in Cortical Cells via Neuronal Circuits** *Front Neurosci* **15**
56. Nishimune A, Isaac JT, Molnar E, Noel J, Nash SR, Tagaya M, Collingridge GL, Nakanishi S, Henley JM (1998) **NSF binding to GluR2 regulates synaptic transmission** *Neuron* **21**:87–97
57. O'Brien RJ, Kamboj S, Ehlers MD, Rosen KR, Fischbach GD, Huganir RL (1998) **Activity-dependent modulation of synaptic AMPA receptor accumulation** *Neuron* **21**:1067–1078
58. Oh MC, Derkach VA (2005) **Dominant role of the GluR2 subunit in regulation of AMPA receptors by CaMKII** *Nat Neurosci* **8**:853–854
59. Ormel L, Stensrud MJ, Bergersen LH, Gundersen V (2012) **VGLUT1 is localized in astrocytic processes in several brain regions** *Glia* **60**:229–238
60. Osten P *et al.* (1998) **The AMPA receptor GluR2 C terminus can mediate a reversible, ATP-dependent interaction with NSF and alpha- and beta-SNAPs** *Neuron* **21**:99–110
61. Pozo K, Goda Y (2010) **Unraveling mechanisms of homeostatic synaptic plasticity** *Neuron* **66**:337–351
62. Riedel D *et al.* (2002) **Rab3D is not required for exocrine exocytosis but for maintenance of normally sized secretory granules** *Mol Cell Biol* **22**:6487–6497
63. Rutherford LC, Nelson SB, Turrigiano GG (1998) **BDNF has opposite effects on the quantal amplitude of pyramidal neuron and interneuron excitatory synapses** *Neuron* **21**:521–530

64. Sanderson JL, Scott JD, Dell'Acqua ML (2018) **Control of Homeostatic Synaptic Plasticity by AKAP-Anchored Kinase and Phosphatase Regulation of Ca(2+)-Permeable AMPA Receptors** *J Neurosci* **38**:2863–2876
65. Schluter OM, Basu J, Sudhof TC, Rosenmund C (2006) **Rab3 superprimes synaptic vesicles for release: implications for short-term synaptic plasticity** *J Neurosci* **26**:1239–1246
66. Schonm JS, van Weering JR, Mohrmann R, Schluter OM, Sudhof TC, de Wit H, Verhage M, Sorensen JB (2010) **Rab3 proteins involved in vesicle biogenesis and priming in embryonic mouse chromaffin cells** *Traffic* **11**:1415–1428
67. Shepherd JD, Rumbaugh G, Wu J, Chowdhury S, Plath N, Kuhl D, Huganir RL, Worley PF (2006) **Arc/Arg3.1 mediates homeostatic synaptic scaling of AMPA receptors** *Neuron* **52**:475–484
68. Silva MM, Rodrigues B, Fernandes J, Santos SD, Carreto L, Santos MAS, Pinheiro P, Carvalho AL (2019) **MicroRNA-186-5p controls GluA2 surface expression and synaptic scaling in hippocampal neurons** *Proc Natl Acad Sci U S A* **116**:5727–5736
69. Smith-Dijak AI, Nassrallah WB, Zhang LY, Geva M, Hayden MR, Raymond LA (2019) **Impairment and Restoration of Homeostatic Plasticity in Cultured Cortical Neurons From a Mouse Model of Huntington Disease** *Front Cell Neurosci* **13**
70. Soares C, Lee KF, Nassrallah W, Beique JC (2013) **Differential subcellular targeting of glutamate receptor subtypes during homeostatic synaptic plasticity** *J Neurosci* **33**:13547–13559
71. Soden ME, Chen L (2010) **Fragile X protein FMRP is required for homeostatic plasticity and regulation of synaptic strength by retinoic acid** *J Neurosci* **30**:16910–16921
72. Song I, Kamboj S, Xia J, Dong H, Liao D, Huganir RL (1998) **Interaction of the N-ethylmaleimide-sensitive factor with AMPA receptors** *Neuron* **21**:393–400
73. Stellwagen D, Malenka RC (2006) **Synaptic scaling mediated by glial TNF- α** *Nature* **440**:1054–1059
74. Stellwagen D, Beattie EC, Seo JY, Malenka RC (2005) **Differential regulation of AMPA receptor and GABA receptor trafficking by tumor necrosis factor- α** *J Neurosci* **25**:3219–3228
75. Suh YH, Terashima A, Petralia RS, Wenthold RJ, Isaac JT, Roche KW, Roche PA (2010) **A neuronal role for SNAP-23 in postsynaptic glutamate receptor trafficking** *Nat Neurosci* **13**:338–343
76. Sun HY, Bartley AF, Dobrunz LE (2009) **Calcium-permeable presynaptic kainate receptors involved in excitatory short-term facilitation onto somatostatin interneurons during natural stimulus patterns** *J Neurophysiol* **101**:1043–1055
77. Sutton MA, Ito HT, Cressy P, Kempf C, Woo JC, Schuman EM (2006) **Miniature neurotransmission stabilizes synaptic function via tonic suppression of local dendritic protein synthesis** *Cell* **125**:785–799
78. Suzuki K, Kim JW, Nosyreva E, Kavalali ET, Monteggia LM (2021) **Convergence of distinct signaling pathways on synaptic scaling to trigger rapid antidepressant action** *Cell Rep* **37**
79. Tataavarty V, Sun Q, Turrigiano GG (2013) **How to scale down postsynaptic strength** *J Neurosci* **33**:13179–13189

80. Thapliyal S, Arendt KL, Lau AG, Chen L (2022) **Retinoic acid-gated BDNF synthesis in neuronal dendrites drives presynaptic homeostatic plasticity** *Elife* **11**
81. Tian M, Xu CS, Montpetit R, Kramer RH (2012) **Rab3A mediates vesicle delivery at photoreceptor ribbon synapses** *J Neurosci* **32**:6931–6936
82. Tononi G, Cirelli C (2014) **Sleep and the price of plasticity: from synaptic and cellular homeostasis to memory consolidation and integration** *Neuron* **81**:12–34
83. Torrado Pacheco A, Bottorff J, Gao Y, Turrigiano GG (2021) **Sleep Promotes Downward Firing Rate Homeostasis** *Neuron* **109**:530–544
84. Trasande CA, Ramirez JM (2007) **Activity deprivation leads to seizures in hippocampal slice cultures: is epilepsy the consequence of homeostatic plasticity?** *J Clin Neurophysiol* **24**:154–164
85. Turrigiano GG, Nelson SB (2004) **Homeostatic plasticity in the developing nervous system** *Nat Rev Neurosci* **5**:97–107
86. Turrigiano GG, Leslie KR, Desai NS, Rutherford LC, Nelson SB (1998) **Activity-dependent scaling of quantal amplitude in neocortical neurons** *Nature* **391**:892–896
87. Wang JKT, Langfelder P, Horvath S, Palazzolo MJ (2017) **Exosomes and Homeostatic Synaptic Plasticity Are Linked to Each other and to Huntington's, Parkinson's, and Other Neurodegenerative Diseases by Database-Enabled Analyses of Comprehensively Curated Datasets** *Front Neurosci* **11**
88. Wang X, Thiagarajan R, Wang Q, Tewolde T, Rich MM, Engisch KL (2008) **Regulation of quantal shape by Rab3A: evidence for a fusion pore-dependent mechanism** *J Physiol* **586**:3949–3962
89. Wang X, Wang Q, Yang S, Bucan M, Rich MM, Engisch KL (2011) **Impaired activity-dependent plasticity of quantal amplitude at the neuromuscular junction of Rab3A deletion and Rab3A earlybird mutant mice** *J Neurosci* **31**:3580–3588
90. Wang X, Li Y, Engisch KL, Nakanishi ST, Dodson SE, Miller GW, Cope TC, Pinter MJ, Rich MM (2005) **Activity-dependent presynaptic regulation of quantal size at the mammalian neuromuscular junction in vivo** *J Neurosci* **25**:343–351
91. Watt AJ, van Rossum MC, MacLeod KM, Nelson SB, Turrigiano GG (2000) **Activity coregulates quantal AMPA and NMDA currents at neocortical synapses** *Neuron* **26**:659–670
92. Weisskopf MG, Castillo PE, Zalutsky RA, Nicoll RA (1994) **Mediation of hippocampal mossy fiber long-term potentiation by cyclic AMP** *Science* **265**:1878–1882
93. Wierenga CJ, Ibata K, Turrigiano GG (2005) **Postsynaptic expression of homeostatic plasticity at neocortical synapses** *J Neurosci* **25**:2895–2905
94. Wilson NR, Kang J, Hueske EV, Leung T, Varoqui H, Murnick JG, Erickson JD, Liu G (2005) **Presynaptic regulation of quantal size by the vesicular glutamate transporter VGLUT1** *J Neurosci* **25**:6221–6234
95. Wu D, Bacaj T, Morishita W, Goswami D, Arendt KL, Xu W, Chen L, Malenka RC, Sudhof TC (2017) **Postsynaptic synaptotagmins mediate AMPA receptor exocytosis during LTP** *Nature* **544**:316–321

96. Wu YK, Hengen KB, Turrigiano GG, Gjorgjieva J (2020) **Homeostatic mechanisms regulate distinct aspects of cortical circuit dynamics** *Proc Natl Acad Sci U S A* **117**:24514–24525
97. Xu X, Pozzo-Miller L (2017) **EEA1 restores homeostatic synaptic plasticity in hippocampal neurons from Rett syndrome mice** *J Physiol* **595**:5699–5712

Editors

Reviewing Editor

Dion Dickman

University of Southern California, Los Angeles, United States of America

Senior Editor

Lu Chen

Stanford University, Stanford, United States of America

Reviewer #1 (Public review):

Koesters and colleagues investigated the role of the small GTPase Rab3A in homeostatic scaling of miniature synaptic transmission in primary mouse cortical cultures using electrophysiology and immunohistochemistry. The major finding is that TTX incubation for 48 hours does not induce an increase in the amplitude of excitatory synaptic miniature events in neuronal cultures derived from Rab3A KO and Rab3A Earlybird mutant mice. NASPM application had comparable effects on mEPSC amplitude in control and after TTX, implying that Ca²⁺-permeable glutamate receptors are unlikely modulated during synaptic scaling. Immunohistochemical analysis revealed no significant changes in GluA2 puncta size, intensity, and integral in control and Rab3A KO cultures. Finally, they provide evidence that loss of Rab3A in neurons, but not astrocytes, blocks homeostatic scaling. Based on these data, the authors propose a model in which neuronal Rab3A is required for homeostatic scaling of synaptic transmission through GluA2-dependent and independent mechanisms.

While the title of the manuscript is mostly supported by data of solid quality, many conclusions, as well as the final model, cannot be derived from the results presented. Importantly, the data do not support that GluA2 levels change upon TTX treatment in control cultures, rendering conclusions regarding Rab3A's role in TTX-dependent GluA2 modulation spurious. Other aspects of the model, such as a Rab3A-dependent release of a tropic factor, cannot be derived from the data.

The following points should be addressed:

- (1) There is no (significant) increase in GluA2 levels (intensity, area, or integral) upon TTX treatment in controls (Fig. 5). Conclusions regarding Rab3A's role in TTX-dependent GluA2 modulation should be revised accordingly. Hence, the data shown in Fig. 4 - 7 do not allow drawing conclusions in the context of Rab3A-dependent GluA2 modulation and scaling.
- (2) The effects of Rab3A on TTX-induced mini frequency modulation remains unclear, because TTX does not induce a change in mini frequency in the Rab3A+/Ebd control (Fig. 2). The respective conclusions should be revised accordingly (l. 427).
- (3) The model is still not supported by the data. In particular, data supporting a negative regulation of Rab3A by APs, Rab3A-dependent release of a tropic factor, or a Rab3A-dependent increase in GluA2 abundance are not presented.

(4) Data points are not overlapping and appear "quantal" in most box plots. How were the data rounded?

<https://doi.org/10.7554/eLife.90261.2.sa3>

Reviewer #2 (Public review):

In the revised manuscript, the authors investigated the role of a presynaptic protein, Rab3A, in the homeostatic synaptic plasticity in cultured cortical neurons. The study was motivated by their previous findings that Rab3A is required for expression of similar homeostatic mechanisms at the neuromuscular junction. The authors first show that untreated WT neurons express homeostatic synaptic plasticity in response to 48h of TTX treatment (upregulation of both mEPSC amplitude and frequency), whereas neurons lacking Rab3A or carrying a dominant negative mutated Rab3A (earlybird) do not. They also demonstrate that only neuronal, but not glial Rab3A is responsible for this defect. Furthermore, they confirm the increased mEPSC amplitudes in WT neurons reflect the addition of GluA2-containing AMPA receptors rather than calcium-permeable ones, as previously reported by multiple labs. However, the increase in mEPSC amplitude is not accompanied by a corresponding upregulation of GluA2 synaptic clusters according to their IHC data (cluster size and intensity trend slightly upwards but not reaching significance). They further show that this modest upward trend is absent in Rab3A KO neurons, and conclude that Rab3A is involved in postsynaptic GluA2 upregulation during homeostatic synaptic plasticity.

When compared to the original version, the authors have done an admirable job in switching to more established ways to assess homeostatic synaptic plasticity by comparing the mean mEPSC amplitude and frequency, which has greatly improved the legibility of the manuscript to the public. Their data in Figures 1,2, and 8 clearly demonstrate that functional Rab3A in cortical neurons is required for the homeostatic regulation of mEPSCs.

However, the authors still have not provided further investigation of the mechanisms behind the role of Rab3A in this form of plasticity, and the revision therefore has added little to the significance of the study. Moreover, the experimental design for the investigation of the mismatch between mEPSC amplitude and GluA2 cluster fluorescence remains questionable, making it difficult to draw any credible conclusions from groups of data that not only look similar to the eye but also show no significance statistically.

A major claim the authors want to make is that Rab3A, although a presynaptic protein, regulates postsynaptic GluA2, and they do this by showing in Figure 5 that the upward trend of GluA2 cluster size and intensity is absent in Rab3A KO neurons. First, it is difficult to convince readers that this upward trend is real in Figures 5B-D without getting more samples. Second, the authors pick GluA2 clusters on the primary dendrites, whereas mEPSC events come from a much larger synapse population (e.g., more distal), therefore it makes sense that these two forms of measurement do not show matching changes, and this caveat could be addressed by sampling more diverse dendritic locations. Without a convincing phenotype in WT neurons, the support for this claim is weak.

Another claim of the authors is that this mismatch between mEPSC amplitude and GluA2 cluster sizes with the same culture suggests there are other factors contributing to the mEPSC amplitude. They do this by comparing results from individual culture dissociations, which greatly suffer from undersampling (Figure 6). In particular, they point out that 2 out of 3 dissociations show "matching" upward trends in mEPSC and GluA2 cluster (figure 6A and 6B) while the third one shows opposite trends, and use this to support their claim. Anyone who has done culture preparation would know the high variability between dissociations, which is why culture data are always pooled for assessment of any population trend. Anything could

have happened to this particular dissociation (culture #3, figure 6C), and drawing conclusion from this single incident does little to support this claim. At least, they should double the dissociation numbers, and their claim would be much more convincing if a similar phenomenon occurs again. Besides, as mentioned above, all these mismatching trends could just be due to sampling differences.

In summary, this study establishes that neuronal Rab3A plays a role in homeostatic synaptic plasticity, but so do a number of other molecules that have been implicated in homeostatic synaptic plasticity in the past two decades (only will grow with the new techniques such as RNAseq). Without going beyond this finding and demonstrating how exactly Rab3A participates in the induction and/or expression of this form of plasticity, or maybe the potential Rab3A-mediated functional and behavioral defects *in vivo*, the contribution of the current study to the field is limited. However, given the presynaptic location of Rab3A, this finding could serve as a starting point for researchers interested in pre-postsynaptic cross-talk during homeostatic plasticity in general.

<https://doi.org/10.7554/eLife.90261.2.sa2>

Reviewer #3 (Public review):

This manuscript presents a number of interesting findings that have the potential to increase our understanding of the mechanism underlying homeostatic synaptic plasticity (HSP). The data broadly support that Rab3A plays a role in HSP, although the site and mechanism of action remain uncertain.

The authors clearly demonstrate the Rab3A plays a role in HSP at excitatory synapses, with substantially less plasticity occurring in the Rab3A KO neurons. There is also no apparent HSP in the Earlybird Rab3A mutation, although baseline synaptic strength seems already elevated. In this context, it is unclear if the plasticity is absent or just occluded by a ceiling effect due the synapses already being strengthened. Occlusion may also occur in the mixed cultures, with Rab3A missing from neurons but not astrocytes. The authors do appropriately discuss both options. There are also differences in genetic background between the Rab3A KO and Earlybird mutants that could also impact the results, which are also noted. The authors have solid data showing that Rab3A is unlikely to be active in astrocytes. Finally, they attempt to study the linkage between synaptic strength during HSP and AMPA receptor trafficking and conclude that trafficking may not be solely responsible for the changes in synaptic strength.

Strengths:

This work adds another player into the mechanisms underlying an important form of synaptic plasticity. The plasticity is likely only reduced, suggesting Rab3A is only partially required and perhaps multiple mechanisms contribute. The authors speculate about some possible novel mechanisms.

However, the conclusions on the partial dissociation of AMPAR trafficking and synaptic response are made from somewhat weaker data. On average, across 3 culture sets, they saw similar magnitude of change in mEPSC amplitude and GluA2 cluster area and integral, but the GluA2 data was not significant. This is likely due to the nature of the datasets. Their imaging method involves only assessing puncta pairs (GluA2/VGlut1) clearly associated with a MAP2 labeled dendrite. This is a small subset of synapses, with usually less than 20 synapses per neuron analyzed (as stated by the authors). The mEPSC recordings will be averaging across several hundred events, which likely represent a hundred or more synapses given reasonable expectations on release probability. It has been reported, in work from this lab as well as by direct monitoring of tagged AMPARs during HSP (Wang, et al., 2019), that

individual synapses are quite variable in their response. So there will almost necessarily be higher variability in the imaging data due to the smaller number of synapses sampled. The overall trends, though, are in alignment with previous data implicating receptor trafficking as the mechanism for HSP. However, the authors go on to evaluate each of the individual cultures, where 2 show similar changes between the mEPSC data and GluA2 clusters, and 1 culture showing little/no change in GluA2 clusters. The n's are very low here, and none of the datasets are significant. They want to conclude for this culture, there was a change in mEPSC amplitude that was not accompanied by a change in GluA2 at synaptic sites. But these data are collected from different coverslips, and due to the low n's, the potential under-sampling of the GluA2 clusters, and neuron-to-neuron variability, it is very hard to distinguish if this apparent difference is a methodological issue rather than a biological one. Much stronger data would be necessary to conclude that additional factors beyond receptor trafficking are required for HSP.

Other questions arise from the NASPM experiments, used to justify looking at GluA2 (and not GluA1) in the immunostaining. First, there is a frequency effect that is unclear in origin. One would expect NASPM to merely block some fraction of the post-synaptic current, and not affect pre-synaptic release or block whole synapses. However the change in frequency seems to argue (as the authors do) that some synapses only have CP-AMPA receptors, while the rest of the synapses have few or none. Another possibility is that there are pre-synaptic NASPM-sensitive receptors that influence release probability. Further, the amplitude data show a strong trend towards smaller amplitude following NASPM treatment (Fig 3B). The p value for both control and TTX neurons was 0.08 - it is very difficult to argue that there is no effect. The decrease on average is larger in the TTX neurons, and some cells show a strong effect. It is possible there is some heterogeneity between neurons on whether GluA1/A2 heteromers or GluA1 homomers are added during HSP. This would impact the weakly supported conclusions about the GluA2 imaging vs mEPSC amplitude data.

Unaddressed issues that would greatly increase the impact of the paper:

(1) Is Rab3A acting pre-synaptically, post-synaptically or both? The authors provide good evidence that Rab3A is acting within neurons and not astrocytes. But where it is acting (pre or post) would aid substantially in understanding its role. They could use sparse knock-down of Rab3A, or simply mix cultures from KO and WT mice (with appropriate tags/labels). The general view in the field has been that HSP is regulated post-synaptically via regulation of AMPAR trafficking, and considerable evidence supports this view. The more support for their suggestion of a pre-synaptic site of control, the better.

(2) Rab3A is also found at inhibitory synapses. It would be very informative to know if HSP at inhibitory synapses is similarly affected. This is particularly relevant as at inhibitory synapses, one expects a removal of GABARs (ie the opposite of whatever is happening at excitatory synapses). If both processes are regulated by Rab3A, this might suggest a role for this protein more upstream in the signaling; an effect only at excitatory synapses would argue for a more specific role just at these synapses.

<https://doi.org/10.7554/eLife.90261.2.sa1>

Author response:

The following is the authors' response to the original reviews.

The detailed, thorough critique provided by the three reviewers is very much appreciated. We believe the manuscript is greatly improved by the changes we have made based on those reviews. The major changes are described below, followed by a point by point response.

Major Changes:

(1) We revised our model (old Fig. 10; new Fig. 9) to keep the explanation focused on the data shown in the current study. Specifically, references to GTP/GDP states of Rab3A and changes in the presynaptic quantum have been removed and the mechanisms depicted are confined to pre- or post-synaptic Rab3A participating in either controlling release of a trophic factor that regulates surface GluA2 receptors (pre- or postsynaptic) or directly affecting fusion of GluA2-receptor containing vesicles (postsynaptic).

(2) We replaced all cumulative density function plots and ratio plots, based on multiple quantile samples per cell, with box plots of cell means. This affects new Figures 1, 2, 3, 5, 6, 7 and 8. All references to “scaling,” “divergent scaling,” or “uniform scaling,” have been removed. New p values for comparison of means are provided above every box plot in Figures 1, 2, 3, 5, 6, 7 and 8. The number of cultures is provided in the figure legends.

(3) We have added frequency to Figures 1, 2 and 8. Frequency values overall are more variable, and the effect of activity blockade less robust, than for mEPSC amplitudes. We have added text indicating that the increase in frequency after activity blockade was significant in neurons from cultures prepared from WT in the Rab3A^{+/−} colony but not cultures prepared from KO mice (Results, lines 143 to 147, new Fig. 1G, H). The TTX-induced increase in frequency was significant in the NASPM experiments before NASPM, but not after NASPM (Results, lines 231 to 233, new Fig. 3, also cultures from WT in Rab3A^{+/−} colony). The homeostatic plasticity effect on frequency did not reach significance in WT on WT glia cultures or

WT on KO glia cultures, possibly due to the variability of frequency, combined with smaller sample sizes (Results, lines 400 to 403, new Fig. 8). In the cultures prepared from WT mice in the Rab3A^{+/−}Ebd colony, there was a trend towards higher frequency after TTX that did not reach statistical significance, and in cultures prepared from mutant mice, the p value was large, suggesting disruption of the effect, which appears to be due to an increase in frequency in untreated cultures, similar to the behavior of mEPSC amplitudes in neurons from mutant mice (Results, lines 161-167). In sum, the effect of activity on frequency requires Rab3A and Ca²⁺-permeable receptors, and is mimicked by the presence of the Rab3A Earlybird mutant. We have also added a discussion of these results (Discussion, lines 427-435).

(4) In the revised manuscript we have added analysis of VGLUT1 levels for the same synaptic sites that we previously analyzed GluA2 levels, and these data are described in Results, lines 344 to 371, and appear in new Table 2. In contrast to previous studies, we did not find any evidence for an increase in VGLUT1 levels after activity blockade. We reviewed those studies to determine whether there might be differences in the experimental details that could explain the lack of effect we observed. In (De Gois et al., 2005), the authors measured mRNA and performed western blots to show increases in VGLUT1 after TTX treatment in older rat cortical cultures (DIV 19). The study performs immunofluorescence imaging of VGLUT1 but only after bicuculline treatment (it decreases), not after TTX treatment. In (Wilson et al.,

2005), the hippocampal cultures are treated with AP5, not TTX, and the VGLUT1 levels in immunofluorescence images are reported relative to synapsin I. That the type of activity blockade matters is illustrated by the failure of Wilson and colleagues to observe a consistent increase in VGLUT1/Synapsin ratio in cultures treated with AMPA receptor blockade (NBQX; supplementary information). These points have been added to the Discussion, lines 436 to 447.)

Reviewer #1:

(1) (model...is not supported by the data), (2) (The analysis of mEPSC data using quantile sampling...), (3) (...statistical analysis of CDFs suffers from n-inflation...), (4) (How does

recording noise and the mEPSC amplitude threshold affect “divergent scaling?” (5) (... justification for the line fits of the ratio data...), (7) (A comparison of p-values between conditions....) and (10) (Was VGLUT intensity altered in the stainings presented in the manuscript?)

The major changes we made, described above, address Reviewer #1’s points. The remaining points are addressed below.

(6) TTX application induces a significant increase in mEPSC amplitude in Rab3A^{-/-} mice in two out of three data sets (Figs. 1 and 9). Hence, the major conclusion that Rab3A is required for homeostatic scaling is only partially supported by the data.

The p values based on CDF comparisons were problematic, but the point we were making is that they were much larger for amplitudes measured in cultures prepared from Rab3A^{-/-} mice (Fig. 1, $p = 0.04$) compared to those from cultures prepared from Rab3A^{+/+} mice (Fig. 1, $p = 4.6 \times 10^{-4}$). Now that we are comparing means, there are no significant TTX-induced effects on mEPSC amplitudes for Rab3A^{-/-} data. However, acknowledging that some increase after activity blockade remains, we describe homeostatic plasticity as being impaired or not significant, rather than abolished, by loss of Rab3A, (Abstract, lines 37 to 39; Results, lines 141 to 143; Discussion, lines 415 to 418).

(8) There is a significant increase in baseline mEPSC amplitude in Rab3A^{Ebd/Ebd} (15 pA) vs. Rab3A^{Ebd/+} (11 pA) cultures, but not in Rab3A^{-/-} (13.6 pA) vs. Rab3A^{+/+} (13.9 pA). Although the nature of scaling was different between Rab3A^{Ebd/Ebd} vs. Rab3A^{Ebd/+} and Rab3A^{Ebd/Ebd} with vs. without TTX, the question arises whether the increase in mEPSC amplitude in Rab3A^{Ebd/Ebd} is Rab3A dependent. Could a Rab3A independent mechanism occlude scaling?

The Reviewer is concerned that the increase in mEPSC amplitude in the presence of the Rab3A point mutant may be through a ‘non-Rab3A’ mechanism (a concern raised by the lack of such effect in cultures from the Rab3A^{-/-} mice), and secondly, that the already large mEPSC cannot be further increased by the homeostatic plasticity mechanism. It must always be considered that a mutant with an altered genetic sequence may bind to novel partners, causing activities that would not be either facilitated or inhibited by the original molecule. We have added this caveat to Results, lines 180 to 186 We added that a number of other manipulations, implicating individual molecules in the homeostatic mechanism, have caused an increase in mEPSC amplitude at baseline, potentially nonspecifically occluding the ability of activity blockade to induce a further increase (Results lines 186 to 189). Still, it is a strong coincidence that the novel activity of the mutant Rab3A would affect mEPSC amplitude, the same characteristic that is affected by activity blockade in a Rab3A dependent manner, a point which we added to Results, lines 189 to 191.

(9) Figure 4: NASPM appears to have a stronger effect on mEPSC frequency in the TTX condition vs. control (-40% vs -15%). A larger sample size might be necessary to draw definitive conclusions on the contribution of Ca²⁺-permeable AMPARs.

Our results, even with the modest sample size of 11 cells, are clear: NASPM does not disrupt the effect of TTX treatment on mEPSC amplitude (new Fig. 3A). It also looks like there is a greater magnitude effect of NASPM on frequency in TTX-treated cells; we note this, but point out that nevertheless, these mEPSCs are not contributing to the increase in mEPSC amplitude (Results, lines 238-241).

(11) The change in GluA2 area or fluorescence intensity upon TTX treatment in controls is modest. How does the GluA2 integral change?

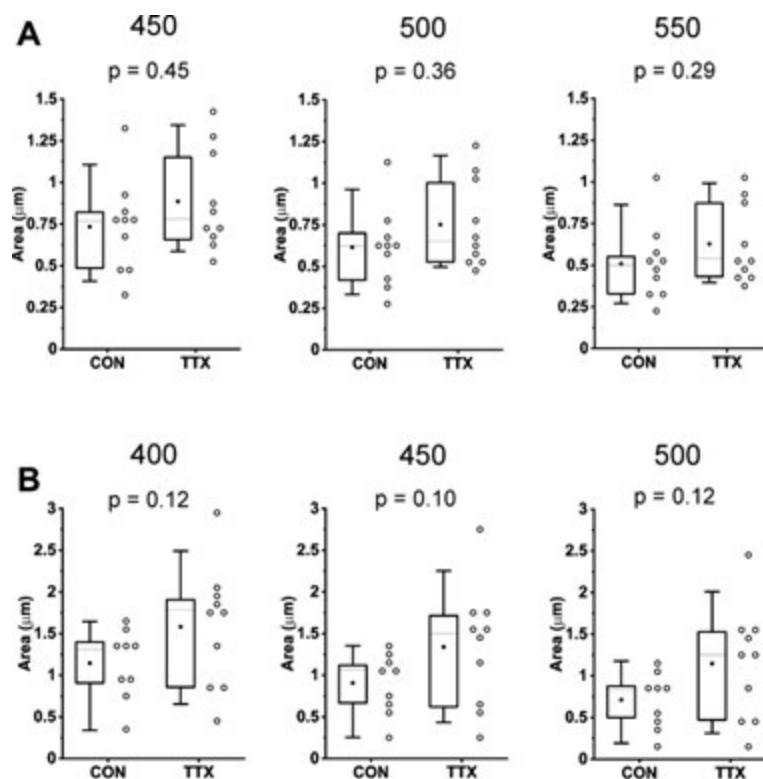
We had reported that GluA2 area showed the most prominent increase following activity blockade, with intensity changing very little. When we examined the integral, it closely matched the change in area. We have added the values for integral to new Fig. 5 D, H; new Fig. 6 A-C; new Fig. 7 A-C and new Table 1 (for GluA2) and new Table 2 (for VGLUT1). These results are described in the text in the following places: Results, lines 289-292; 298-299; 311-319; 328-324). For VGLUT1, both area and intensity changed modestly, and the integral appeared to be a combination of the two, being higher in magnitude and resulting in smaller p values than either area or intensity (Results, lines 344-348; 353-359; new Table 2).

(12) The quantitative comparison between physiology and microscopy data is problematic. The authors report a mismatch in ratio values between the smallest mEPSC amplitudes and the smallest GluA2 receptor cluster sizes (l. 464; Figure 8). Is this comparison affected by the fluorescence intensity threshold? What was the rationale for a threshold of 400 a.u. or 450 a.u.? How does this threshold compare to the mEPSC threshold of 3 pA.

This concern is partially addressed by no longer comparing the rank ordered mEPSC amplitudes with the rank ordered GluA2 receptor characteristics. We had used multiple thresholds in the event that an experiment was not analyzable with the chosen threshold (this in fact happened for VGLUT1, see end of this paragraph). We created box plots of the mean GluA2 receptor cluster size, intensity and integral, for experiments in which we used all three thresholds, to determine if the effect of activity blockade was different depending on which threshold was applied, and found that there was no obvious difference in the results (Author response image 1). Nevertheless, since there is no need to use a different threshold for any of the 6 experiments (3 WT and 3KO), for new Figures 5, 6 and 7 we used the same threshold for all data, 450; described in Methods, lines 746 to 749. For VGLUT1 levels, it was necessary to use a different threshold for Rab3A^{+/+} Culture #1 (400), but a threshold of 200 for the other five experiments (Methods, lines 751-757). The VGLUT1 immunofluorescent sites in Culture #1 had higher levels overall, and the low threshold caused the entire AOI to be counted as the synapse, which clearly included background levels outside of the synaptic site. Conversely, to use a threshold of 400 on the other experiments meant that the synaptic site found by the automated measurement tool was much smaller than what was visible by eye. In our judgement it would have been meaningless to adhere to a single threshold for VGLUT1 data.

Author response image 1.

Using different thresholds does not substantially alter GluA2 receptor cluster size data. A) Rab3A^{+/+} Culture #1, size data for three different thresholds, depicted above each graph. B) Rab3A^{+/+} Culture #2, size data for three different thresholds, depicted above each graph. Note scale bar in A is different from B, to highlight differences for different thresholds. (Culture #3 was only analyzed with 450 threshold).



The conclusion that an increase in AMPAR levels is not fully responsible for the observed mEPSC increase is mainly based on the rank-order analysis of GluA2 intensity, yielding a slope of ~ 0.9 . There are several points to consider here: (i) GluA2 fluorescence intensity did increase on average, as did GluA2 cluster size.

(ii) The increase in GluA2 cluster size is very similar to the increase in mEPSC amplitude (each approx. 1820%). (iii) Are there any reports that fluorescence intensity values are linearly reporting mEPSC amplitudes (in this system)? Antibody labelling efficiency, and false negatives of mEPSC recordings may influence the results. The latter was already noted by the authors.

Our comparison between mEPSC amplitude and GluA2 receptor cluster characteristics has been reexamined in the revised version using means rather than rank-ordered data in rank-order plots or ratio plots. Importantly, all of these methods revealed that in one out of three WT cultures (Culture #3) GluA2 receptor cluster size (old Fig. 8, old Table 1; new Fig. 6, new Table 1), intensity and integral (new Fig. 6, new Table 1) values decreased following activity blockade while in the same culture, mEPSC amplitudes increased. It is based on this lack of correspondence that we conclude that increases in mEPSC amplitude are not fully explained by increases in GluA2 receptors, and suggest there may be other contributors. These points are made in the Abstract (lines 108-110); Results (lines 319 to 326; 330337; 341-343) and the Discussion (lines 472 to 474). To our knowledge, there are not any reports that quantitatively compare receptor levels (area, intensity or integrals) to mEPSC amplitudes in the same cultures. We examined the comparisons very closely for 5 studies that used TTX to block activity and examined receptor levels using confocal imaging at identified synapses (Hou et al., 2008; Ibata et al., 2008; Jakawich et al., 2010a; Xu and Pozzo-Miller, 2017; Dubes et al., 2022). We were specifically looking for whether the receptor data were more variable than the mEPSC amplitude data, as we found. However, for 4 of the studies, sample sizes were very different so that we cannot simply compare the p values. Below is a table of the comparisons.

Author response table 1.

Reference	Sample size	% change	p value
Hou et al., 2008			
Receptor data	40,000-50,000 sites	40%	< 0.05
mEPSC data	15 cells	95%	< 0.05
Ibata et al., 2008			
Receptor data	360 puncta from 20 cells	Not provided but < 45%	< 0.01
mEPSC data	10-11 cells	45%	< 0.03
Jakawich et al., 2010			
Receptor data	61-82 cells	30%, GluA1; 40%, GluA2	< 0.05
mEPSC data	8-16 cells	33%	< 0.05
Xu and Pozo-Miller 2017			
Receptor data	13-14 cells	60%	<0.05
mEPSC data	9-10 cells	60%	<0.01
Dubes et al., 2022			
Receptor data	26-33 cells	30%	0.0014
mEPSC data	65-68 cells	27%	< 0.0001

In Xu 2017 the sample sizes are close enough that we feel comfortable concluding that the receptor data were slightly more variable ($p < 0.05$) than mEPSC data ($p < 0.01$) but recognize that it is speculative to say our finding has been confirmed. A discussion of these articles is in Discussion, lines 456-474.

(iv) It is not entirely clear if their imaging experiments will sample from all synapses. Other AMPAR subtypes than GluA2 could contribute, as could kainite or NMDA receptors.

While our imaging data only examined GluA2, we used the application of NASPM to demonstrate Ca^{2+} -permeable receptors did not contribute quantitatively to the increase in mEPSC amplitude following TTX treatment. Since GluA3 and GluA4 are also Ca^{2+} -permeable, the findings in new Figure 3 (old Fig. 4) likely rule out these receptors as well. There are also reports that Kainate receptors are Ca^{2+} -permeable and blocked by NASPM (Koike et al., 1997; Sun et al., 2009), suggesting the NASPM experiment also rules out the contribution of Kainate receptors. Finally, given our recording conditions, which included normal magnesium levels in the extracellular solution as well as TTX to block action-potential evoked synaptic transmission, NMDA receptors would not be available to contribute currents to our recordings due to block by magnesium ions at resting V_m . These points have been added to the Methods section, lines 617 to 677 (NMDA); 687-694 (Ca^{2+} -permeable AMPA receptors and Kainate receptors).

*Furthermore, the statement “complete lack of correspondence of TTX/CON ratios” is not supported by the data presented (l. 515ff). First, under the assumption that no scaling occurs in *Rab3A*^{-/-}, the TTX/CON ratios show a 20-30% change, which indicates the variation of this readout. Second, the two examples shown in Figure 8 for *Rab3A*^{+/-} are*

actually quite similar (culture #1 and #2, particularly when ignoring the leftmost section of the data, which is heavily affected by the raw values approaching zero.

We are no longer presenting ratio plots in the revised manuscript, so we do not base our conclusion that mEPSC amplitude data is not always corresponding to GluA2 receptor data on the difference in behavior of TTX/CON ratio values, but only on the difference in direction of the TTX effect in one out of three cultures. We agree with the reviewer that the ratio plots are much more sensitive to differences between control and treated values than the rank order plot, and we feel these differences are important, for example, there is still a homeostatic increase in the Rab3A^{-/-} cultures, and the effect is still divergent rather than uniform. But the comparison of ratio data will be presented elsewhere.

(13) Figure 7A: TTX CDF was shifted to smaller mEPSC amplitude values in Rab3A^{-/-} cultures. How can this be explained?

While this result is most obvious in CDF plots, we still observe a trend towards smaller mEPSC amplitudes after TTX treatment in two of three individual cultures prepared from Rab3A^{-/-} mice when comparing means (new Fig. 7, Table 1) which did not reach statistical significance for the pooled data (new Fig. 5, new Table 1). There was not any evidence of this decrease in the larger data set (new Fig. 1) nor for Rab3A^{-/-} neurons on Rab3A^{+/+} glia (new Fig. 8). Given that this effect is not consistent, we did not comment on it in the revised manuscript. It may be that there is a non-Rab3A-dependent mechanism that results in a decrease in mEPSC amplitude after activity blockade, which normally pulls down the magnitude of the activity-dependent increase typically observed. But studying this second component would be difficult given its magnitude and inconsistent presentation.

Reviewer #1 (Recommendations For the Authors):

(1) Abstract, last sentence: The conclusion of the present manuscript should be primarily based on the results presented. At present, it is mainly based on a previous publication by the authors.

We have revised the last sentence to reflect actual findings of the current study (Abstract, lines 47 to 49).

(2) Line 55: “neurodevelopmental”

This phrase has been removed.

(3) Line 56: “AMPAergic” should be replaced by AMPAR-mediated

This sentence was removed when all references to “scaling” were removed; no other instances of “AMPAergic” are present.

(4) Figure 9: The use of BioRender should be disclosed in the Figure Legend.

We used BioRender in new Figures 3, 7 and 8, and now acknowledge BioRender in those figure legends.

(5) Figure legends and results: The number of cultures should be indicated for each comparison.

Number of cultures has been added to the figure legends.

(6) Line 289: A comparison of *p*-values between conditions does not allow any meaningful conclusions.

Agreed, therefore we have removed CDFs and the KS test comparison *p* values. All comparisons in the revised manuscript are for cell means.

(7) Line 623ff: The argument referring to NMJ data is weak, given that different types of receptors are involved.

We still think it is valid to point out that Rab3A is required for the increase in mEPSC at the NMJ but that ACh receptors do not increase (Discussion, lines 522 to 525). We are not saying that postsynaptic receptors do not contribute in cortical cultures, only that there could be another Rab3A-dependent mechanism that also affects mEPSC amplitude.

(8) Plotting data points outside of the ranges should be avoided (e.g., Fig. 2Giii, 7F).

These two figures are no longer present in the revised manuscript. In revising figures, we made sure no other plots have data points outside of the ranges.

(9) The rationale for investigating Rab3AEbd/Ebd remains elusive and should be described.

A rationale for investigating Rab3AEbd/Ebd is that if the results are similar to the KO, it strengthens the evidence for Rab3A being involved in homeostatic synaptic plasticity. In addition, since its phenotype of early awakening was stronger than that demonstrated in Rab3A KO mice (Kapfhamer et al., 2002), it was possible we would see a more robust effect. These points have been added to the Results, lines 118 to 126.

(10) Figures 3 and 4, as well as Figure 5 and 6 could be merged.

In the revised version, Figure 3 has been eliminated since its main point was a difference in scaling behavior. Figure 4 has been expanded to include a model of how NASPM could reduce frequency (new Fig. 3.) Images of the pyramidal cell body have been added to Figure 5 (new Fig. 4), and Figure 6 has been completely revised and now includes pooled data for both Rab3A^{+/+} and Rab3A^{-/-} cultures, for mEPSC amplitude, GluA2 receptor cluster size, intensity and integral.

(11) Figure 5: The legend refers to MAP2, but this is not indicated in the figure.

MAP2 has now been added to the labels for each image and described in the figure legend (new Fig. 4).

Reviewer #2:

Technical concerns:

(1) The culture condition is questionable. The authors saw no NMDAR current present during spontaneous recordings, which is worrisome since NMDARs should be active in cultures with normal network activity (Watt et al., 2000; Sutton et al., 2006). It is important to ensure there is enough spiking activity before doing any activity manipulation. Similarly it is also unknown whether spiking activity is normal in Rab3AKO/Ebd neurons.

In the studies cited by the reviewer, NMDA currents were detected under experimental conditions in which magnesium was removed. In our recordings, we have normal magnesium (1.3 mM) and also TTX, which prevents the necessary depolarization to allow inward current through NMDA receptors. This point has been added to our Methods, lines 674 to 677. We acknowledge we do not know the level of spiking in cultures prepared from Rab3A^{+/+}, Rab3A^{-/-} or Rab3A_Ebd/Ebd_ mice. Given the similar mEPSC amplitude for untreated cultures from WT and KO studies, we think it unlikely that activity was low in the latter, but it remains a possibility for untreated cultures from Rab3A_Ebd/Ebd_ mice, where mEPSC amplitude was increased. These points are added to the Methods, lines 615 to 622.

(2) Selection of mEPSC events is not conducted in an unbiased manner. Manually selecting events is insufficient for cumulative distribution analysis, where small biases could skew the entire distribution. Since the authors claim their ratio plot is a better method to detect the uniformity of scaling than the well-established rank-order plot, it is important to use an unbiased population to substantiate this claim.

We no longer include any cumulative distributions or ratio plot analysis in the revised version. We have added the following text to Methods, lines 703 to 720:

“MiniAnalysis selects many false positives with the automated feature when a small threshold amplitude value is employed, due to random fluctuations in noise, so manual re-evaluation of the automated process is necessary to eliminate false positives. If the threshold value is set high, there are few false positives but small amplitude events that visually are clearly mEPSCs are missed, and manual re-evaluation is necessary to add back false negatives or the population ends up biased towards large mEPSC amplitudes. As soon as there is a manual step, bias is introduced. Interestingly, a manual reevaluation step was applied in a recent study that describes their process as ‘unbiased (Wu et al., 2020). In sum, we do not believe it is currently possible to perform a completely unbiased detection process. A fully manual detection process means that the same criterion (“does this *look* like an mEPSC?”) is applied to all events, not just the false positives, or the false negatives, which prevents the bias from being primarily at one end or the other of the range of mEPSC amplitudes. It is important to note that when performing the MiniAnalysis process, the researcher did not know whether a record was from an untreated cell or a TTX-treated cell.”

(3) Immunohistochemistry data analysis is problematic. The authors only labeled dendrites without doing cell-fills to look at morphology, so it is questionable how they differentiate branches from pyramidal neurons and interneurons. Since glutamatergic synapse on these two types of neuron scale in the opposite directions, it is crucial to show that only pyramidal neurons are included for analysis.

We identified neurons with a pyramidal shape and a prominent primary dendrite at 60x magnification without the zoom feature. This should have been made clear in the description of imaging. We have added an image of the two selected cells to our figure of dendrites (old Fig. 5, new Fig. 4), and described this process in the Methods, lines 736 to 739, and Results, lines 246 to 253. Given the morphology of the neurons selected it is highly unlikely that the dendrites we analyzed came from interneurons.

Conceptual Concerns

The only novel finding here is the implicated role for Rab3A in synaptic scaling, but insights into mechanisms behind this observation are lacking. The authors claim that Rab3A likely regulates scaling from the presynaptic side, yet there is no direct evidence from data presented. In its current form, this study's contribution to the field is very limited.

We have demonstrated that loss of Rab3A and expression of a Rab3A point mutant disrupt homeostatic plasticity of mEPSC amplitudes, and that in the absence of Rab3A, the increase in GluA2 receptors at synaptic sites is abolished. Further, we show that this effect cannot be through release of a factor, like TNF α , from astrocytes. In the new version, we add the finding that VGLUT1 is not increased after activity blockade, ruling out this presynaptic factor as a contributor to homeostatic increases in mEPSC amplitude. We show for the first time by examining mEPSC amplitudes and GluA2 receptors in the same cultures that the increases in GluA2 receptors are not as consistent as the increases in mEPSC amplitude, suggesting the possibility of another contributor to homeostatic increases in mEPSC amplitude. We first proposed this idea in our previous study of Rab3A-dependent homeostatic increases in mEPSC amplitudes at the mouse neuromuscular junction. In sum, we dispute that there is only one novel finding and that we have no insights into mechanism. We acknowledge that we have no direct evidence for regulation from the presynaptic side, and have removed this claim from the revised manuscript. We have retained the Discussion of potential mechanisms affecting the presynaptic quantum and evidence that Rab3A is implicated in these mechanisms (vesicle size, fusion pore kinetics; Discussion, lines 537 to 563). One way to directly show that the amount of transmitter released for an mEPSC has been modified after activity blockade is to demonstrate that a fast off-rate antagonist has become less effective at inhibiting mEPSCs (because the increased glutamate released out competes it; see (Liu et al., 1999) and (Wilson et al., 2005) for example experiments). This set of experiments is underway but will take more time than originally expected, because we are finding surprisingly large decreases in frequency, possibly the result of mEPSCs with very low glutamate concentration that are completely inhibited by the dose used. Once mEPSCs are lost, it is difficult to compare the mEPSC amplitude before and after application of the antagonist. Therefore we intend to include this experiment in a future report, once we determine the reason for the frequency reduction, or, can find a dose where this does not occur.

(1) Their major argument for this is that homeostatic effects on mEPSC amplitudes and GluA2 cluster sizes do not match. This is inconsistent with reports from multiple labs showing that upscaling of mEPSC amplitude and GluA2 accumulation occur side by side during scaling (Ibata et al., 2008; Pozo et al., 2012; Tan et al., 2015; Silva et al., 2019). Further, because the acquisition and quantification methods for mEPSC recordings and immunohistochemistry imaging are entirely different (each with its own limitations in signal detection), it is not convincing that the lack of proportional changes must signify a presynaptic component.

Within the analyses in the revised manuscript, which are now based only on comparison of cell/dendrite means, we find a very good match in the magnitude of increase for the pooled data of mEPSC amplitudes and GluA2 receptor cluster sizes (+19.7% and +20.0% respectively; new Table 1). However, when looking at individual cultures, we had one of three WT cultures in which mEPSC amplitude increased 17.2% but GluA2 cluster size decreased 9.5%. This result suggests that while activity blockade does lead to an increase in GluA2 receptors after activity blockade, the effect is more variable than that for mEPSC amplitude. We went back to published studies to see if this has been previously observed, but found that it was difficult to compare because the sample sizes were different for the two characteristics (see Author response table 1). We included these particular 5 studies because they use the same treatment (TTX), examine receptors using imaging of identified synaptic sites, and record mEPSCs in their cultures (although the authors do not indicate that imaging and recordings are done simultaneously on the same cultures.) Only one of the studies listed by the Reviewer is in our group (Ibata et al., 2008). The study by (Tan et al., 2015) uses western blots to measure receptors; the study by (Silva et al., 2019) blocks activity using a combination of AMPA and NMDA receptor blockers; the study by (Pozo et al., 2012) correlates mEPSC amplitude changes with imaging but not in response to activity blockade, instead for changing the expression of

GluA2. While it may seem like splitting hairs to reject studies that use other treatment protocols, there is ample evidence that the mechanisms of homeostatic plasticity depend on how activity was altered, see the following studies for several examples of this (Sutton et al., 2006; Soden and Chen, 2010; Fong et al., 2015). A discussion of the 5 articles we selected is in the revised manuscript, Discussion, lines 456 to 474. In sum, we provide evidence that activity blockade is associated with an overall increase in GluA2 receptors; what we propose is that this increase, being more variable, does not fully explain the increase in mEPSC amplitude. However, we acknowledge that the disparity could be explained by the differences in limitations of the two methods (Discussion, lines 469-472).

(2) The authors also speculate in the discussion that presynaptic Rab3A could be interacting with retrograde BDNF signaling to regulate postsynaptic AMPARs. Without data showing Rab3A-dependent presynaptic changes after TTX treatment, this argument is not compelling. In this retrograde pathway, BDNF is synthesized in and released from dendrites (Jakawich et al., 2010b; Thapliyal et al., 2022), and it is entirely possible for postsynaptic Rab3A to interfere with this process cell-autonomously.

We have added the information that Rab3A could control BDNF from the postsynaptic cell and included the two references provided by the reviewer, Discussion, lines 517 to 518. We have added new evidence, recently published, that the Rab3 family has been shown to regulate targeting of EGF receptors to rafts (among other plasma membrane molecules), with Rab3A itself clearly present in nonneuronal cells (Diaz-Rohrer et al., 2023) (added to Discussion, lines 509 to 515).

(3) The authors propose that a change in AMPAR subunit composition from GluA2-containing ones to GluA1 homomers may account for the distinct changes in mEPSC amplitudes and GluA2 clusters. However, their data from the NASPM wash-in experiments clearly show that the GluA1 homomer contributions have not changed before and after TTX treatment.

We have revised this section in the Discussion, lines 534 to 536, to clarify that any change due to GluA1 homomers should have been detectable by a greater ability of NASPM to reverse the TTX-induced increase.

Reviewer #2 (Recommendations for the Authors):

For authors to have more convincing arguments in general, they will need to clarify/improve certain details in their data collection by addressing the above technical concerns. Additionally, the authors should design experiments to test whether Rab3A regulates scaling from pre- or post-synaptic site. For example, they could sparsely knock out Rab3A in WT neurons to test the postsynaptic possibility. On the other hand, their argument for a presynaptic role would be much more compelling if they could show whether there are clear functional changes such as in vesicle sizes and release probability in the presynaptic terminal of Rab3AKO neurons.

An important next step is to identify whether Rab3A is acting pre- or post-synaptically (Discussion, lines 572 to 573), but these experiments will be undertaken in the future. It would not add much to simply show vesicle size is altered in the KO (and we do not necessarily expect this since mEPSC amplitude is normal in the KO). It will be very difficult to establish that vesicle size is changing with activity blockade and that this change is prevented in the Rab3A KO, because we are looking for a ~25% increase in vesicle volume, which would correspond to a ~7.5% increase in diameter. Finally, we do not believe demonstrating changes in release probability tell us anything about a presynaptic role for Rab3A in regulating the size of the presynaptic quantum.

Reviewer #3 (Public Review)

Weaknesses: However, the rather strong conclusions on the dissociation of AMPAR trafficking and synaptic response are made from somewhat weaker data. The key issue is the GluA2 immunostaining in comparison with the mEPSC recordings. Their imaging method involves only assessing puncta clearly associated with a MAP2 labeled dendrite. This is a small subset of synapses, judging from the sample micrographs (Fig. 5). To my knowledge, this is a new and unvalidated approach that could represent a particular subset of synapses not representative of the synapses contributing to the mEPSC change (they are also sampling different neurons for the two measurements; an additional unknown detail is how far from the cell body were the analyzed dendrites for immunostaining.) While the authors acknowledge that a sampling issue could explain the data, they still use this data to draw strong conclusions about the lack of AMPAR trafficking contribution to the mEPSC amplitude change. This apparent difference may be a methodological issue rather than a biological one, and at this point it is impossible to differentiate these. It will unfortunately be difficult to validate their approach. Perhaps if they were to drive NMDA-dependent LTD or chemLTP, and show alignment of the imaging and ephys, that would help. More helpful would be recordings and imaging from the same neurons but this is challenging. Sampling from identified synapses would of course be ideal, perhaps from 2P uncaging combined with SEP-labeled AMPARs, but this is more challenging still. But without data to validate the method, it seems unwarranted to make such strong conclusions such as that AMPAR trafficking does not underlie the increase in mEPSC amplitude, given the previous data supporting such a model.

In the new version, we soften our conclusion regarding the mismatch between GluA2 receptor levels and mEPSC amplitudes, now only stating that receptors may not be the sole contributor to the TTX effect on mEPSC amplitude (Discussion, lines 472 to 474). With our analysis in the new version focusing on comparisons of cell means, the GluA2 receptor cluster size and the mEPSC amplitude data match well in magnitude for the data pooled across the 3 matched cultures (20.0% and 19.7%, respectively, see new Table 1). However, in one of the three cultures the direction of change for GluA2 receptors is opposite that of mEPSC amplitudes (Table 1, Culture #3, -9.5% vs +17.2%, respectively).

It is unlikely that the lack of matching of homeostatic plasticity in one culture, but very good matching in two other cultures, can be explained by an unvalidated focus on puncta associated with MAP2 positive dendrites. We chose to restrict analysis of synaptic GluA2 receptors to the primary dendrite in order to reduce variability, reasoning that we are always measuring synapses for an excitatory pyramidal neuron, synapses that are relatively close to the cell body, on the consistently identifiable primary dendrite. We measured how far this was for the two cells depicted in old Figure 5 (new Fig. 4). Because we always used the 5X zoom window which is a set length, and positioned it within ~10 microns of the cell body, these cells give a ball park estimate for the usual distances. For the untreated cell, the average distance from the cell body was $38.5 \pm 2.8 \mu\text{m}$; for the TTX-treated cell, it was $42.4 \pm 3.2 \mu\text{m}$ ($p = 0.35$, KruskalWallis test). We have added these values to the Results, lines 270 to 274.

We did not mean to propose that AMPA receptor levels do not contribute at all to mEPSC amplitude, and we acknowledge there are clear cases where the two characteristics change in parallel (for example, in the study cited by Reviewer #2, (Pozo et al., 2012), increases in GluA2 receptors due to exogenous expression are closely matched by increases in mEPSC amplitudes.) What our matched culture experiments demonstrate is that in the case of TTX treatment, both GluA2 receptors and mEPSC amplitudes increase on average, but sometimes mEPSC amplitudes can increase in the absence of an increase in GluA2 receptors (Culture #3, Rab3A+/+ cultures), and sometimes mEPSC amplitudes do not increase even though GluA2 receptor levels do increase (Culture #3, Rab3A-/- cultures). Therefore, it would not add

anything to our argument to examine receptors and mEPSCs in NMDA-dependent LTP, a different plasticity paradigm in which changes in receptors and mEPSCs may more closely align. It has been demonstrated that mEPSCs of widely varying amplitude can be recorded from a single synaptic site (Liu and Tsien, 1995), so we would need to measure a large sample of individual synapse recordings to detect a modest shift in average values due to activity blockade. In addition, it would be essential to express fluorescent AMPA receptors in order to correlate receptor levels in the same cells we record from (or at the same synapses). And yet, even after these heroics, one is still left with the issue that the two methods, electrophysiology and fluorescent imaging, have distinct limitations and sources of variability that may obscure any true quantitative correlation.

Other questions arise from the NASPM experiments, used to justify looking at GluA2 (and not GluA1) in the immunostaining. First, there is a frequency effect that is quite unclear in origin. One would expect NASPM to merely block some fraction of the post-synaptic current, and not affect pre-synaptic release or block whole synapses. It is also unclear why the authors argue this proves that NASPM was at an effective concentration (lines 399-400). Further, the amplitude data show a strong trend towards smaller amplitude. The p value for both control and TTX neurons was 0.08 – it is very difficult to argue that there is no effect. And the decrease is larger in the TTX neurons. Considering the strong claims for a presynaptic locus and the use of this data to justify only looking at GluA2 by immunostaining, these data do not offer much support of the conclusions. Between the sampling issues and perhaps looking at the wrong GluA subunit, it seems premature to argue that trafficking is not a contributor to the mEPSC amplitude change, especially given the substantial support for that hypothesis. Further, even if trafficking is not the major contributor, there could be shifts in conductance (perhaps due to regulation of auxiliary subunits) that does not necessitate a pre-synaptic locus. While the authors are free to hypothesize such a mechanism, it would be prudent to acknowledge other options and explanations.

We have created a model cartoon to explain how NASPM could reduce mEPSC frequency (new Fig. 3D). mEPSCs that arise from a synaptic site that has only Ca²⁺-permeable AMPA receptors will be completely blocked by NASPM, if the NASPM concentration is maximal. The reason we conclude that we have sufficient NASPM reaching the cells is that the frequency is decreased, as expected if there are synaptic sites with only Ca²⁺-permeable AMPA receptors. We previously were not clear that there is an effect of NASPM on mEPSC amplitude, although it did not reach statistical significance (new Fig. 3B). Where there is no effect is on the TTX-induced increase in mEPSC amplitude, which remains after the acute NASPM application (new Fig. 3A). We have revised the description of these findings in Results, lines 220 to 241. In reviewing the literature further, we could find no previous studies demonstrating an increase in conductance in GluA2 or Ca²⁺-impermeable receptors, only in GluA1 homomers. In other words, any conductance change would have been due to a change in GluA1 homomers, and should have been visible as a disruption of the homeostatic plasticity by NASPM application. We have added text to Results, lines 211 to 217; 236-241; Discussion, lines 420 to 422; 526-536 and Methods, lines 685 to 695 regarding this point.

The frequency data are missing from the paper, with the exception of the NASPM dataset. The mEPSC frequencies should be reported for all experiments, particularly given that Rab3A is generally viewed as a pre-synaptic protein regulating release. Also, in the NASPM experiments, the average frequency is much higher in the TTX treated cultures. Is this statistically above control values?

This comment is addressed by the major change #3, above.

Unaddressed issues that would greatly increase the impact of the paper:

(1) Is Rab3A activity pre-synaptically, post-synaptically or both. The authors provide good evidence that Rab3A is acting within neurons and not astrocytes. But where is it acting (pre or post) would aid substantially in understanding its role (and particularly the hypothesized and somewhat novel idea that the amount of glutamate released per vesicle is altered in HSP). They could use sparse knockdown of Rab3A, or simply mix cultures from KO and WT mice (with appropriate tags/labels). The general view in the field has been that HSP is regulated post-synaptically via regulation of AMPAR trafficking, and considerable evidence supports this view. The more support for their suggestion of a pre-synaptic site of control, the better.

This is similar to the request of Reviewer #2, Recommendations to the Authors. An important next step is to identify whether Rab3A is working pre- or postsynaptically. However, it is possible that it is acting pre-synaptically to anterogradely regulate trafficking of AMPAR, as we have depicted in our model, new Fig. 9. To demonstrate that the presynaptic quantum is being altered, we would need to show that vesicle size is increased, or the amount of transmitter being released during an mEPSC is increased after activity blockade. To that end, we are currently performing experiments using a fast off-rate antagonist. As described above in response to Reviewer #2's Conceptual Concerns, we find dramatic decreases in frequency not explained by the 30-60% inhibition observed for the largest amplitude mEPSCs, which suggests the possibility that small mEPSCs are more sensitive than large mEPSCs and therefore may have less transmitter. Due to these complexities and the delay while we test other antagonists to see if the effect is specific to fast-off rate antagonists, we are not including these results here.

(2) Rab3A is also found at inhibitory synapses. It would be very informative to know if HSP at inhibitory synapses is similarly affected. This is particularly relevant as at inhibitory synapses, one expects a removal of GABARs and/or a decrease of GABA-packaging in vesicles (ie the opposite of whatever is happening at excitatory synapses.). If both processes are regulated by Rab3A, this might suggest a role for this protein more upstream in the signaling, an effect only at excitatory synapses would argue for a more specific role just at these synapses.

It will be important to determine if homeostatic synaptic plasticity at inhibitory synapses on excitatory neurons is sensitive to Rab3A deletion, especially in light of the fact that unlike many of the other molecules implicated in homeostatic increases in mEPSCs, Rab3A is not a molecule known to be selective for glutamate receptor trafficking (in contrast to Arc/Arg3.1 or GRIP1, for example). Such a study would warrant its own publication.

Reviewer #3 (Recommendations for the Authors):

There are a number of minor points or suggestions for the authors:

Is RIM1 part of this pathway (or expected to be)? Some discussion of this would be nice.

RIM, Rab3-interacting molecule, has been implicated at the drosophila neuromuscular junction in a presynaptic form of homeostatic synaptic plasticity in which evoked release is increased after block of postsynaptic receptors (Muller et al., 2012), a plasticity that also requires Rab3-GAP (Muller et al., 2011). To our knowledge there is no evidence that RIM is involved in the homeostatic plasticity of mEPSC amplitude after activity blockade by TTX. The Rim1a KO does not have a change in mEPSC amplitude relative to WT (Calakos et al., 2004), but that is not unexpected given the normal mEPSC amplitude in neurons from cultures prepared from Rab3A^{-/-} mice in the current study. It would be interesting to look at homeostatic plasticity in cortical cultures prepared from Rim1a or other RIM deletion mice, but we have not added these points to the revised manuscript since there are a number of

directions one could go in attempting to define the molecular pathway and we feel it is more important to discuss the potential location of action and physiological mechanisms.

Is the Earlybird mutation a GOF? More information about this mutation would help.

We have added a description of how the Earlybird mutation was identified, in a screen for rest:activity mutants (Results, lines 118 to 123). Rab3A Earlybird mice have a shortened circadian period, shifting their wake cycle earlier and earlier. When Rab3A deletion mice were tested in the same activity raster plot measurements, the shift was smaller than that for the Earlybird mutant, suggesting the possibility that it is a dominant negative mutation.

The high K used in the NASPM experiments seems a bit unusual. Have the authors done high K/no drug controls to see if this affects the synapses in any way?

We used the high K based on previous studies that indicated the blocking effect of the Ca²⁺-permeable receptor blockers was use dependent (Herlitze et al., 1993; Iino et al., 1996; Koike et al., 1997). We reasoned that a modest depolarization would increase the frequency of AMPA receptor mEPSCs and allow access of the NASPM. We have added this point to the Methods, lines 695 to 708.

The NASPM experiments do not show that GluA1 does not contribute (line 401), only that GluA1 homomers are not contributing (much – see above). GluA1/A2 heteromers are quite likely involved. Also, the SEM is missing from the WT pre/post NASPM data.

Imaging of GluA2-positive sites will not distinguish between GluA2 homomers and GluA2-GluA1 heteromers, so we have added this clarification to Results, lines 242 to 246. We have remade the NASPM pre-post line plots so that the mean values and error bars are more visible (new Fig. 3B, C).

It seems odd to speculate based on non-significant findings (line 650-1), with lower significance ($p = 0.11$) than findings being dismissed in the paper (NASPM on mEPSC amplitude; $p = 0.08$).

We did not mean to dismiss the effect of NASPM on mEPSC amplitude (new Fig. 3B), rather, we dismiss the effect of NASPM on the homeostatic increase in mEPSC amplitude caused by TTX treatment (new Fig. 3A). We have emphasized this distinction in Results, lines 223 to 225, and Discussion, lines 420 to 422, as well as adding that the stronger effect of NASPM on frequency after TTX treatment suggests an activity-dependent increase in the number of synapses expressing only Ca²⁺ permeable homomers (Results, lines 236 to 241; Discussion, lines 431 to 435).

Fig. 4 could be labeled better (to make it clear that B is amplitude and C is freq from the same cells).

Fig. 4 has been revised—now the amplitude and frequency plots from the same condition (new Fig. 3, B, C; CON or TTX) are in a vertical line and the figure legend states that the frequency data are from the same cells as in Fig. 3A.

The raw amplitude data seems a bit hidden in the inset panels – I would suggest these data are at least as important as the cumulative distributions in the main panel. Maybe re-organizing the figures would help.

We have removed all cumulative distributions, rank order plots, and ratio plots. The box plots are now full size in new Figures 1, 2, 5, 6, 7 and 8.

I'm not sure I would argue in the paper that 12 cells a day is a limiting issue for experiments. It doesn't add anything and doesn't seem like that high a barrier. It is fine to just say it is difficult and therefore there is a limited amount of data meeting the criteria.

We have removed the comment regarding difficulty.

Calakos N, Schoch S, Sudhof TC, Malenka RC (2004) Multiple roles for the active zone protein RIM1alpha in late stages of neurotransmitter release. *Neuron* 42:889-896.

De Gois S, Schafer MK, Defamie N, Chen C, Ricci A, Weihe E, Varoqui H, Erickson JD (2005) Homeostatic scaling of vesicular glutamate and GABA transporter expression in rat neocortical circuits. *J Neurosci* 25:7121-7133.

Diaz-Rohrer B, Castello-Serrano I, Chan SH, Wang HY, Shurer CR, Levental KR, Levental I (2023) Rab3 mediates a pathway for endocytic sorting and plasma membrane recycling of ordered microdomains. *Proc Natl Acad Sci U S A* 120:e2207461120.

Dubes S, Soula A, Benquet S, Tessier B, Poujol C, Favereaux A, Thoumine O, Letellier M (2022) miR-124-dependent tagging of synapses by synaptopodin enables input-specific homeostatic plasticity. *EMBO J* 41:e109012.

Fong MF, Newman JP, Potter SM, Wenner P (2015) Upward synaptic scaling is dependent on neurotransmission rather than spiking. *Nat Commun* 6:6339.

Herlitze S, Raditsch M, Ruppersberg JP, Jahn W, Monyer H, Schoepfer R, Witzemann V (1993) Argitoxin detects molecular differences in AMPA receptor channels. *Neuron* 10:1131-1140.

Hou Q, Zhang D, Jarzylo L, Huganir RL, Man HY (2008) Homeostatic regulation of AMPA receptor expression at single hippocampal synapses. *Proc Natl Acad Sci U S A* 105:775-780.

Ibata K, Sun Q, Turrigiano GG (2008) Rapid synaptic scaling induced by changes in postsynaptic firing. *Neuron* 57:819-826.

Iino M, Koike M, Isa T, Ozawa S (1996) Voltage-dependent blockage of Ca(2+)-permeable AMPA receptors by joro spider toxin in cultured rat hippocampal neurones. *J Physiol* 496 (Pt 2):431437.

Jakowich SK, Neely RM, Djakovic SN, Patrick GN, Sutton MA (2010a) An essential postsynaptic role for the ubiquitin proteasome system in slow homeostatic synaptic plasticity in cultured hippocampal neurons. *Neuroscience* 171:1016-1031.

Jakowich SK, Nasser HB, Strong MJ, McCartney AJ, Perez AS, Rakesh N, Carruthers CJ, Sutton MA (2010b) Local presynaptic activity gates homeostatic changes in presynaptic function driven by dendritic BDNF synthesis. *Neuron* 68:1143-1158.

Kapfhamer D, Valladares O, Sun Y, Nolan PM, Rux JJ, Arnold SE, Veasey SC, Bucan M (2002) Mutations in Rab3a alter circadian period and homeostatic response to sleep loss in the mouse. *Nat Genet* 32:290-295.

Koike M, Iino M, Ozawa S (1997) Blocking effect of 1-naphthyl acetyl spermine on Ca(2+)-permeable AMPA receptors in cultured rat hippocampal neurons. *Neurosci Res* 29:27-36.

Liu G, Tsien RW (1995) Properties of synaptic transmission at single hippocampal synaptic boutons. *Nature* 375:404-408.

Liu G, Choi S, Tsien RW (1999) Variability of neurotransmitter concentration and nonsaturation of postsynaptic AMPA receptors at synapses in hippocampal cultures and

slices. *Neuron* 22:395409.

Muller M, Pym EC, Tong A, Davis GW (2011) Rab3-GAP controls the progression of synaptic homeostasis at a late stage of vesicle release. *Neuron* 69:749-762.

Muller M, Liu KS, Sigrist SJ, Davis GW (2012) RIM controls homeostatic plasticity through modulation of the readily-releasable vesicle pool. *J Neurosci* 32:16574-16585.

Pozo K, Cingolani LA, Bassani S, Laurent F, Passafaro M, Goda Y (2012) beta3 integrin interacts directly with GluA2 AMPA receptor subunit and regulates AMPA receptor expression in hippocampal neurons. *Proc Natl Acad Sci U S A* 109:1323-1328.

Silva MM, Rodrigues B, Fernandes J, Santos SD, Carreto L, Santos MAS, Pinheiro P, Carvalho AL (2019) MicroRNA-186-5p controls GluA2 surface expression and synaptic scaling in hippocampal neurons. *Proc Natl Acad Sci U S A* 116:5727-5736.

Soden ME, Chen L (2010) Fragile X protein FMRP is required for homeostatic plasticity and regulation of synaptic strength by retinoic acid. *J Neurosci* 30:16910-16921.

Sun HY, Bartley AF, Dobrunz LE (2009) Calcium-permeable presynaptic kainate receptors involved in excitatory short-term facilitation onto somatostatin interneurons during natural stimulus patterns. *J Neurophysiol* 101:1043-1055.

Sutton MA, Ito HT, Cressy P, Kempf C, Woo JC, Schuman EM (2006) Miniature neurotransmission stabilizes synaptic function via tonic suppression of local dendritic protein synthesis. *Cell* 125:785-799.

Tan HL, Queenan BN, Hugarir RL (2015) GRIP1 is required for homeostatic regulation of AMPAR trafficking. *Proc Natl Acad Sci U S A* 112:10026-10031.

Thapliyal S, Arendt KL, Lau AG, Chen L (2022) Retinoic acid-gated BDNF synthesis in neuronal dendrites drives presynaptic homeostatic plasticity. *Elife* 11.

Wilson NR, Kang J, Hueske EV, Leung T, Varoqui H, Murnick JG, Erickson JD, Liu G (2005) Presynaptic regulation of quantal size by the vesicular glutamate transporter VGLUT1. *J Neurosci* 25:6221-6234.

Wu YK, Hengen KB, Turrigiano GG, Gjorgjieva J (2020) Homeostatic mechanisms regulate distinct aspects of cortical circuit dynamics. *Proc Natl Acad Sci U S A* 117:24514-24525.

Xu X, Pozzo-Miller L (2017) EEA1 restores homeostatic synaptic plasticity in hippocampal neurons from Rett syndrome mice. *J Physiol* 595:5699-5712.

<https://doi.org/10.7554/eLife.90261.2.sa0>